

Modeling Systemic Risk and Tail Dependence in an Emerging Stock Market: Evidence from Vietnam

Vo Tran Huong Giang

Student ID: 2211345002

Supervisor: Nguyen Thi Hoang Anh

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Abstract

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1 Introduction

1.1 Research Rationale

Financial crises affect equity markets not only through losses at the individual-firm level, but also through the synchronization of negative returns across firms and sectors. When downside movements become concentrated, the benefits of diversification may decline precisely during periods of market stress. In classical portfolio theory, diversification reduces risk when asset returns are imperfectly correlated (Markowitz, 1952). However, this argument is based on average co-movement and may be less informative when dependence changes across the return distribution. Assets that appear only moderately related under normal conditions may experience stronger joint losses in the lower tail. Therefore, an important issue for the Vietnamese equity market is not only the level of stock-return volatility, but also whether extreme downside movements become clustered in ways that increase market-wide fragility.

This issue is particularly relevant for emerging equity markets, where liquidity constraints, concentrated investor behaviour, information frictions, and shifts in global risk appetite can intensify downside co-movement during periods of stress. In such markets, average correlation may understate the extent of joint downside risk. C. Zhou and Qin (2021) show that asymmetric tail dependence is important for measuring extreme equity risk and improving international portfolio performance, while Duong and Huynh (2020a) document tail dependence across ASEAN-6 equity markets. These findings suggest that extreme co-movement is a relevant feature of emerging and regional equity markets. For Vietnam, where the stock market has become increasingly important for capital allocation, household investment, and corporate financing, understanding lower-tail dependence is therefore relevant not only for portfolio risk measurement, but also for financial-stability analysis.

Vietnam's equity market provides an appropriate setting for examining this issue. HOSE is the country's main listed equity market and includes many of the largest firms in banking, real estate, materials, consumer sectors, and industrials. These sectors are linked to credit cycles, property-market conditions, trade exposure, and domestic liquidity. During periods of stress, shocks may be transmitted across sectors through valuation effects, investor sentiment, margin lending, and portfolio rebalancing. For instance, declines in real estate stocks may affect banks through credit exposure and collateral values, while stress in materials or energy stocks may reflect broader concerns about trade conditions, input costs, or global demand. The relevant question is whether these sectoral linkages remain contained or develop into broader channels of systemic downside-risk transmission during market downturns.

Conventional correlation-based measures may not fully capture this form of risk. Pearson correlation measures average linear co-movement across the full return distribution, whereas financial contagion is often asymmetric and state-dependent. Two stocks may not move closely together during normal periods, but may still experience joint extreme losses during market

stress. Lower-tail dependence is designed to capture this type of relationship. From a risk-management perspective, it measures whether two assets are likely to experience extreme negative returns together. From an economic perspective, it reflects the weakening of diversification benefits, the concentration of downside risk, and the potential transmission of shocks across sectors. McNeil, Frey, and Embrechts (2015a) emphasize the importance of tail dependence in quantitative risk management, as extreme losses are the states in which standard distributional assumptions may become less reliable.

Lower-tail dependence is also relevant for regulatory and macroprudential analysis. Systemic risk is not limited to the risk of a large individual firm, but concerns the extent to which distress in one part of the system contributes to broader market instability. Measures such as CoVaR, SRISK, and marginal expected shortfall have been developed to identify institutions whose distress has market-wide implications (Adrian and Brunnermeier, 2016; Acharya et al., 2017a; Brownlees and Engle, 2017). Although these measures are commonly applied to financial institutions, the same logic can be extended to equity-market networks. Stocks or sectors that occupy central positions in downside-dependence networks may act as important channels of crash-risk transmission. For Vietnam, this perspective is useful because market supervision often emphasizes volatility, liquidity, and price movements, while the structure of downside interdependence may receive less attention.

A tail-dependence network approach can therefore provide additional information for investors, asset managers, exchanges, and regulators. For investors, it helps assess whether sectoral diversification remains effective during crisis periods. For asset managers, it supports portfolio construction that accounts for joint downside risk rather than relying only on average correlation. For regulators, it can help identify which sectors form the systemic core of the market and whether this core changes across different shock regimes. This is especially relevant for Vietnam, where market stress may originate from different sources, including global trade disruptions, pandemic shocks, domestic credit conditions, and external financial spillovers. A framework that identifies sectoral crash transmission can help distinguish between persistent sources of systemic risk and regime-specific risk rotations.

Accordingly, this study examines lower-tail dependence and sectoral systemic risk on HOSE. The central motivation is that market crises are not only periods of falling prices, but also periods in which the dependence structure of the market may change. If conventional correlation does not adequately identify stocks and sectors that jointly decline in the lower tail, then risk monitoring based only on average co-movement may underestimate systemic vulnerability. By focusing on lower-tail dependence, this study aims to provide a more suitable measure of crash co-movement and to identify the sectors that contribute most to downside-risk transmission in the Vietnamese stock market.

1.2 Research Background and Gap

The literature on systemic risk has developed several approaches to measure how distress in one institution or market segment can affect the broader financial system. Adrian and Brunnermeier (2016) introduce CoVaR, which measures the value-at-risk of the financial system conditional on the distress of a specific institution. Acharya et al. (2017a) propose systemic expected shortfall, while Brownlees and Engle (2017) develop SRISK as a conditional capital shortfall measure. These contributions are important because they shift the focus from the risk of individual institutions to their contribution to system-wide risk. However, these measures are mainly designed for financial institutions and capital shortfall analysis. For stock-market applications, particularly in emerging markets, an additional perspective is needed to examine how downside dependence among listed firms and sectors contributes to systemic market risk.

A related stream of research examines contagion and spillovers across equity markets. Bekaert et al. (2014) study equity market contagion during the global financial crisis and show that crisis transmission cannot be fully explained by normal-period market linkages. Diebold and Yilmaz (2012) develop spillover measures that are widely used to quantify connectedness across financial variables. In the Vietnamese context, T. T. A. Tran (2023) examine risk spillovers from the U.S. stock market to Vietnam using VAR-DCC-GARCH and quantile regression, while V. C. Nguyen and Thu Thuy Nguyen (2022) analyze dependence between Chinese and Vietnamese stock markets during the COVID-19 period. These studies show that Vietnam is exposed to external financial shocks. However, much of the existing evidence is cross-market or aggregate in nature, and therefore provides limited evidence on how risk is transmitted within the domestic stock market across individual stocks and sectors.

Vietnam-specific research has increasingly examined systemic risk, dependence, and financial connectedness. Vu and D. K. Tran (2019) study systemic risk in the Vietnamese stock market, while P. A. Nguyen, N. P. Nguyen, and H. M. D. Le (2023) construct a Financial Risk Meter to assess systemic risk and financial network conditions in Vietnam. More recently, A. T. H. Nguyen and T. T. Le (2025) examine financial connectedness among Vietnamese banks, insurance firms, and securities companies using the Diebold and Yilmaz (2012) network framework. They document a high level of interconnectedness among Vietnamese financial institutions, especially after the COVID-19 pandemic and recent episodes of bank runs and market stress. Their findings also show close connectedness between the banking and securities sectors, with banks traditionally acting as primary shock transmitters and securities firms becoming increasingly relevant in the post-2020 period. Sectoral evidence is also developing. Dang, N. T. Nguyen, and Vo (2023) analyze sectoral volatility spillovers and their determinants in Vietnam, while Mensi et al. (2024) investigate extreme quantile connectedness between oil and Vietnamese stock sectors. These studies indicate that financial risk transmission in Vietnam is heterogeneous across institutions and sectors, and that connectedness tends to increase during stress periods. Nevertheless, less attention has been given to stock-level lower-tail dependence

networks that identify which HOSE-listed stocks and sectors become central to downside market transmission.

Another relevant body of research focuses on copulas and tail dependence. Copula methods are useful because they separate marginal distributions from dependence structures and allow asymmetric dependence to be modeled more flexibly than linear correlation (Sklar, 1959; Nelsen, 2006; Joe, 2014). This is particularly relevant in financial markets, where joint downside movements may differ from average co-movement or joint upside movements. Duong and Huynh (2020a) show that ASEAN equity markets exhibit tail dependence, while Li and Rose (2009a) emphasize that emerging markets may display stronger dependence during large joint losses. In the Vietnamese context, Thu Thuy Nguyen, V. C. Nguyen, and Tran Trong Nguyen (2020) apply a copula approach to oil and gas-related firms, and T. A. Le and Dao (2021) use copulas for portfolio optimization under mean-CVaR simulation on the Vietnamese stock exchange. These studies support the usefulness of copulas for modeling nonlinear and tail dependence. However, many existing applications focus on dependence estimation or portfolio optimization, rather than translating lower-tail dependence into systemic importance rankings of stocks and sectors.

Network analysis provides a way to move from pairwise dependence to systemic interpretation. Financial networks can identify central nodes, clusters, and transmission channels that may not be visible from a full dependence matrix. Tumminello et al. (2005a) propose the Planar Maximally Filtered Graph as a method for extracting informative links from complex systems while preserving richer structure than a minimum spanning tree. Wen, Yang, and W. Zhou (2018) show that tail-dependence networks of global stock markets differ from Pearson correlation networks and can reveal different structures of financial risk. D. Wang and Huang (2021) develop centrality-based systemic importance measures using lower-tail dependence networks and principal component analysis. These studies suggest that combining copulas with filtered networks can improve systemic-risk measurement. However, there remains limited evidence applying this approach to Vietnam's stock market at the stock and sector level.

Taken together, the literature suggests three related gaps. First, existing Vietnamese studies have shown that systemic risk, volatility spillovers, and external contagion are relevant, but they provide limited stock-level evidence on domestic downside transmission channels. Second, copula studies have shown that tail dependence exists in emerging and regional markets, but often do not translate pairwise dependence into systemic importance for further examination of the channels through which crash risk propagates across sectors and key market participants. Third, network studies show that centrality can identify systemically important nodes, but this approach has not been widely applied to lower-tail dependence among HOSE-listed stocks. What remains underexplored is whether lower-tail dependence can identify the domestic sectoral channels through which crash risk is transmitted in Vietnam's equity market, and whether these channels differ from those implied by conventional correlation. This distinction is important because a stock may be highly correlated with the market under normal conditions without being a central

crash transmitter, while another stock may exhibit moderate average correlation but stronger lower-tail connectedness during stress periods.

This study makes three contributions. First, it contributes to systemic-risk measurement by constructing a tail-dependence-based systemic importance index for HOSE-listed stocks. This allows systemic importance to be interpreted not only through size or volatility, but also through a stock's position in the downside dependence network. Second, it contributes to the literature on dependence modeling by comparing lower-tail copula dependence with Pearson correlation in the context of crisis co-movement. This comparison assesses whether tail dependence provides economically distinct information relative to conventional correlation. Third, it contributes to sectoral financial-stability analysis in Vietnam by identifying which sectors form persistent or regime-specific sources of downside-risk transmission.

The study is therefore positioned at the intersection of systemic-risk measurement, tail-dependence modeling, and financial network analysis. Its primary contribution lies in identifying which stocks and sectors matter most for market-wide downside risk. Tail-dependence modeling provides the dependence structure, while network analysis translates that structure into systemic importance. By focusing on HOSE-listed stocks from 2016 to 2025, the study links the methodology to economically meaningful market regimes, including normal periods, trade-related shocks, pandemic stress, domestic funding stress, and recent tariff-related uncertainty. This design allows the analysis to examine whether sectoral systemic risk remains stable over time or changes according to the nature of each shock.

1.3 Research Aim, Objectives and Questions

The aim of this study is to examine how lower-tail dependence shapes sectoral systemic risk and crash transmission in the Vietnamese stock market. Rather than treating market risk solely as a function of individual stock volatility, this study focuses on the structure of joint downside movements among HOSE-listed stocks. The central premise is that systemic risk becomes more relevant when losses are not evenly distributed across firms, but are concentrated within specific sectors or transmitted through highly connected stocks. Accordingly, the study investigates whether lower-tail dependence reveals a structure of market fragility that may not be fully captured by conventional correlation-based measures.

More specifically, the study addresses three related issues. First, it examines whether extreme downside co-movement exists among HOSE-listed stocks and whether this dependence is more concentrated within sectors or across sectors. This issue is important because a sectorally clustered lower-tail structure would imply that some industries are more exposed to simultaneous losses and may provide limited diversification benefits during market stress. Second, the study identifies which stocks and sectors occupy central positions in the downside dependence network, and whether their systemic importance changes across market regimes. This allows the analysis to distinguish between stocks that are individually volatile and stocks that are

systemically important because they are strongly connected to other stocks during adverse market conditions. Third, the study evaluates whether a Clayton lower-tail dependence framework provides additional information relative to a Pearson correlation framework in explaining crisis-period losses and sectoral risk transmission.

The first objective is to assess the degree and sectoral structure of lower-tail dependence among HOSE-listed stocks. This objective examines whether downside co-movement is a broad market-wide feature or whether it is concentrated in specific intra-sector and cross-sector channels. In economic terms, it identifies where diversification benefits are most likely to weaken during market downturns.

The second objective is to measure the systemic importance of individual stocks and sectors under downside market conditions, and to examine whether this importance changes across market regimes. This objective moves beyond pairwise dependence by focusing on each stock's position within the broader lower-tail network. A stock is considered more systemically important if it is strongly connected to other stocks in the lower-tail dependence network and therefore has greater potential to participate in, or transmit, crash risk. By comparing normal periods with crisis episodes such as the Trade War, COVID-19, the Bond Shock, and tariff-related shocks, this objective also assesses whether systemic sectors remain stable over time or rotate according to the nature of each shock.

The third objective is to compare tail-dependence-based systemic risk measurement with conventional correlation-based measurement. This objective evaluates whether Pearson correlation, which captures average linear co-movement, overlooks downside linkages that become more visible in the lower tail. This comparison is relevant because investors and regulators often rely on correlation-based tools, although crisis-period dependence may be asymmetric, nonlinear, and state-dependent.

Based on these objectives, the study addresses the following research questions:

- RQ1.** What is the degree and sectoral structure of lower-tail dependence among HOSE-listed stocks?
- RQ2.** Which stocks and sectors contribute most to market-wide downside risk and crash transmission on HOSE, and does their systemic importance change across crisis regimes?
- RQ3.** Does lower-tail copula dependence capture downside co-movement more effectively than conventional Pearson correlation?

1.4 Methodology Overview

The study uses daily stock return data for HOSE-listed firms from January 2016 to December 2025. The sample is selected to reflect sectoral market-capitalization weights, while also accounting for firm-level liquidity and data availability. Daily logarithmic returns are first filtered

using a univariate GJR-GARCH(1,1) model with skewed- t innovations. This step controls for serial dependence, volatility clustering, and conditional heteroskedasticity before dependence estimation. The subsequent analysis is therefore based on standardized innovations rather than raw return volatility.

The main empirical framework focuses on lower-tail dependence. Pairwise Clayton copulas are estimated from the standardized innovations to obtain lower-tail dependence coefficients among HOSE-listed stocks. The Clayton copula is used because it captures the tendency of two assets to experience joint extreme negative outcomes, making it suitable for analyzing crash co-movement and downside contagion. These pairwise lower-tail dependence estimates are then used to construct a Planar Maximally Filtered Graph, denoted as $PMFG_{tail}$, which extracts the most informative connections from the full dependence matrix while preserving the network structure needed for systemic-risk analysis.

To identify systemically important stocks within the lower-tail network, four centrality measures are computed: weighted degree, betweenness, closeness, and eigenvector centrality. These measures capture different aspects of a stock's role in the downside transmission network, including its total connectedness, bridging position, network accessibility, and connection to other central nodes. Principal component analysis is then used to combine these centrality measures into a tail-based systemic importance index, denoted as SII_{tail} . This index provides the main measure of each stock's contribution to lower-tail systemic risk.

The empirical analysis proceeds in three stages. First, the study examines the level and distribution of pairwise lower-tail dependence, with particular attention to intra-sector and cross-sector patterns. This stage assesses whether downside co-movement is broadly distributed across the market or concentrated in specific sectoral channels. Second, the analysis ranks stocks and sectors by SII_{tail} across the full sample and across market regimes. This allows the study to identify persistent systemic cores as well as regime-specific rotations in crash-risk transmission. Third, a Pearson-correlation-based network and corresponding SII_{corr} are constructed as a benchmark for conventional dependence measurement. The study compares this benchmark with the lower-tail framework using network overlap, rank correlation, and crisis validation regressions. This final stage evaluates whether lower-tail systemic importance provides additional explanatory power for crisis-period losses relative to standard correlation-based measurement.

1.5 Scope and Limitations

The scope of this study is limited to HOSE-listed stocks over the period 2016–2025. This focus is appropriate because HOSE is Vietnam's main equity trading venue and includes many of the country's most liquid and systemically relevant listed firms. The sample period covers both normal market conditions and several stress episodes, including the US–China Trade War, the COVID-19 crisis, and the 2025 Trump Tariff shock. This setting allows the study to examine

whether lower-tail dependence remains stable across market regimes or varies under different types of shocks.

Several limitations should be acknowledged. First, the study relies on pairwise copula estimation, which captures bilateral lower-tail dependence but does not model the full high-dimensional dependence structure of the market. Second, the Clayton copula focuses specifically on lower-tail co-movement and does not capture upper-tail dependence. This choice is consistent with the study's focus on downside systemic risk, although it means that upside co-movement is outside the scope of the analysis. Third, the systemic importance index is based on stock-price information and therefore does not directly capture balance-sheet linkages, ownership networks, or credit exposures. Fourth, the analysis is limited to HOSE-listed stocks and may not be fully generalizable to HNX, UPCoM, or unlisted financial exposures. Finally, the Trump Tariff window is relatively short, so the results for this sub-period should be interpreted with caution.

Overall, this study uses lower-tail dependence as a framework for examining systemic downside risk in Vietnam's stock market. By comparing tail-based and correlation-based networks, the analysis assesses whether conventional dependence measures are sufficient for crisis-period monitoring, or whether a downside-specific framework provides additional information for identifying channels of crash transmission.

2 Literature Review

2.1 Systemic Risk, Financial Contagion, and Market Interconnectedness

Systemic risk refers to the possibility that distress in one part of the financial system can spread to other institutions, markets, or sectors, thereby impairing the stability of the broader system. Unlike individual risk, which concerns the vulnerability of a single firm or asset, systemic risk is fundamentally relational. It emerges from interdependence, common exposures, and the structure through which shocks are transmitted. This distinction is central to modern financial stability analysis because a firm or sector may appear manageable in isolation but become systemically relevant when its distress affects other parts of the market.

The global financial crisis shifted the literature from a microprudential view of risk toward a system-wide perspective. Earlier models of financial contagion show that linkages among institutions can be stabilizing under moderate shocks because they allow risk sharing and liquidity redistribution. Allen and Gale (2000), for example, argue that a more complete financial network may improve resilience by diversifying liquidity risk. However, subsequent studies emphasize that the stabilizing role of interconnectedness is conditional. Acemoglu, Ozdaglar and Tahbaz-Salehi (2015) show that dense financial networks may absorb small shocks but become fragile when shocks exceed a critical threshold. In such cases, the same linkages that support diversification can become channels of contagion.

This dual role of interconnectedness is particularly relevant for equity markets. Listed firms

are connected not only through direct balance-sheet exposures, but also through investor portfolios, sectoral fundamentals, funding conditions, market sentiment, and liquidity constraints. During normal periods, these connections may support price discovery and risk sharing. During stress periods, however, they can amplify losses when investors rebalance simultaneously, margin constraints tighten, or sentiment deteriorates across related sectors. Financial contagion therefore does not require direct contractual links; it can also occur through market-based channels reflected in stock returns and volatility.

Several systemic risk measures have been developed to capture the contribution of individual institutions to system-wide distress. Adrian and Brunnermeier (2016) propose CoVaR to measure the value-at-risk of the financial system conditional on the distress of a particular institution. Acharya et al. (2017a) develop systemic expected shortfall, while Brownlees and Engle (2017) introduce SRISK as a conditional capital shortfall measure. These measures are important because they shift the analytical focus from the risk of an individual entity to its contribution to aggregate risk. Nevertheless, many of these frameworks are designed primarily for financial institutions and capital adequacy analysis. For stock market applications, especially at the sectoral level, there remains a need for approaches that reveal how market-based shocks are transmitted through dependence structures among listed firms.

The connectedness literature provides one response to this problem. Diebold and Yilmaz (2012) develop a variance-decomposition-based framework to measure spillovers across financial variables and to represent these spillovers as networks. This approach has been widely used because it captures both total and directional connectedness, allowing researchers to distinguish between net transmitters and receivers of shocks. However, connectedness based on volatility or average return dynamics may still fail to capture the asymmetric nature of crisis transmission. Market stress is often concentrated in the lower tail of the return distribution, where losses become synchronized and diversification breaks down. This motivates the need to examine systemic risk not only through general connectedness, but also through lower-tail dependence.

2.2 Tail Dependence and the Failure of Diversification During Crises

Portfolio diversification is built on the idea that imperfectly correlated assets can reduce aggregate risk. Markowitz (1952) formalizes this principle by showing that portfolio risk depends not only on individual asset variances but also on the covariance structure among assets. In normal market conditions, this framework provides a powerful foundation for portfolio allocation. However, its practical effectiveness weakens when the dependence structure among assets changes during crisis periods. Assets that appear weakly or moderately related in average conditions may experience strong joint losses under extreme downside states.

Tail dependence directly addresses this problem. It measures the tendency of assets to experience extreme outcomes together. Lower-tail dependence is especially important in equity markets because it captures the probability that assets jointly fall into extreme negative return

states. Economically, this matters because the social and financial costs of crises arise not from ordinary volatility, but from the clustering of severe losses. When lower-tail dependence is high, diversification benefits decline precisely when investors most need them. This makes lower-tail dependence relevant for portfolio risk management, stress testing, systemic risk monitoring, and macroprudential policy.

The limitation of linear correlation is therefore not merely statistical. Pearson correlation summarizes average linear co-movement over the entire distribution. It does not distinguish between normal-period co-movement and crisis-period co-movement, nor does it capture asymmetric dependence between gains and losses. A pair of stocks may have moderate correlation in normal times but still exhibit strong lower-tail dependence during market downturns. Conversely, a high correlation pair may not necessarily be the most relevant transmitter of crash risk. This distinction is central to downside risk analysis because systemic crises are tail events rather than average events.

The empirical literature provides considerable evidence that tail dependence is relevant for financial markets. Studies on international and emerging markets show that dependence often becomes stronger during periods of stress, especially in the lower tail. C. Nguyen, Bhatti, and Henry (2017), for instance, examine tail dependence between the US stock market and the Vietnamese and Chinese stock markets and find evidence of stronger left-tail dependence between the US and Vietnam, suggesting that these markets are more prone to crashing together than booming together during extreme events. Duong and Huynh (2020a) also document tail dependence among ASEAN-6 equity markets, confirming that extreme co-movement is an important feature of regional emerging markets. These findings imply that conventional dependence measures may understate the vulnerability of emerging markets to joint downside shocks.

For the present study, the relevance of lower-tail dependence is twofold. First, it provides a direct measure of downside co-movement among HOSE-listed stocks. Second, it allows the paper to examine whether such downside dependence is sectorally structured. If lower-tail dependence is concentrated within or between particular sectors, then market-wide risk is not evenly distributed. Instead, some sectors may act as important channels through which crash risk is transmitted. This makes lower-tail dependence an economically meaningful object of study rather than a purely statistical feature of returns.

2.3 Copula-Based Dependence Modelling

2.3.1 Nonlinear and Asymmetric Dependence in Financial Markets

Financial returns typically exhibit non-normality, volatility clustering, skewness, and fat tails. These characteristics make dependence modelling difficult because the joint behaviour of returns cannot be fully described by linear correlation or multivariate normal assumptions. Copula theory offers a flexible solution by separating marginal distributions from the dependence struc-

ture. Since Sklar's theorem states that any multivariate joint distribution can be decomposed into marginal distributions and a copula function, copulas allow researchers to model the dependence between variables independently from their individual distributional properties.

This feature is especially useful in financial markets. Stock returns often differ in volatility, skewness, and tail thickness across firms and sectors. A copula-based approach allows each return series to be filtered and transformed into a comparable marginal scale before estimating dependence. As a result, the dependence structure can be analyzed without imposing identical distributional assumptions across all assets. This is important for emerging stock markets, where firm-level return distributions are often heterogeneous due to differences in liquidity, information quality, ownership structure, and sectoral exposure.

Copulas are also suitable for modelling nonlinear and asymmetric dependence. Unlike Pearson correlation, which captures only average linear association, copulas can describe how dependence behaves in different parts of the joint distribution. This matters because financial markets often exhibit stronger dependence during downturns than upturns. In other words, assets may not rise together as strongly as they fall together. Such asymmetry has direct implications for downside risk management because investors and regulators are more concerned with joint losses than joint gains.

Previous studies have applied copula models to examine tail dependence, contagion, and portfolio risk. Patton (2012) highlights the usefulness of copulas in financial econometrics, particularly for modelling asymmetric dependence. McNeil, Frey, and Embrechts (2015a) also emphasize the importance of tail dependence for quantitative risk management. In emerging market settings, copula-based studies show that downside dependence may be substantial even when conventional correlations appear moderate. Therefore, copulas provide an appropriate framework for this study because the paper focuses specifically on joint downside movements rather than average co-movement.

2.3.2 Lower-Tail Dependence and the Clayton Copula

Among copula families, the Clayton copula is particularly relevant for modelling lower-tail dependence. Its main property is that it captures dependence in the lower tail while implying no upper-tail dependence. This feature is appropriate for the present study because the research question concerns crash co-movement and market-wide downside risk, rather than co-movement during positive market states. In this context, the absence of upper-tail dependence is not a weakness but a deliberate modelling choice. It focuses the analysis on the part of the distribution that is most relevant for systemic risk.

The Clayton lower-tail dependence coefficient provides an interpretable measure of joint downside risk. A higher coefficient indicates that two stocks are more likely to experience extreme negative returns together. This differs from Pearson correlation, which treats upside and downside co-movement symmetrically. By focusing on the lower tail, the Clayton framework can identify pairs of stocks that may not appear highly connected under average conditions but

become strongly linked during market stress.

This distinction is important for sectoral systemic risk analysis. In equity markets, a sector may not always be central in normal trading conditions, but it may become a systemic transmitter during crisis periods if its stocks share strong lower-tail dependence with the broader market. For example, banking and real estate stocks may be closely linked through credit exposure, collateral values, and investor expectations, while materials and energy stocks may become important during trade or commodity-related shocks. A lower-tail copula framework is therefore well suited to identifying the sectors through which crash risk becomes concentrated.

The choice of Clayton copula also aligns with recent studies that combine lower-tail dependence with systemic importance measurement. D. Wang and Huang (2021) construct lower-tail dependence networks using the Clayton copula and Planar Maximally Filtered Graphs, then derive a centrality-based systemic importance index using principal component analysis. Their approach shows that lower-tail dependence networks can capture cross-sectional and time-series variation in systemic importance. This supports the present study's use of Clayton-based lower-tail dependence as the foundation for identifying systemically important stocks and sectors in Vietnam.

2.4 Financial Networks and Systemic Importance Measurement

Pairwise dependence measures are informative but difficult to interpret when the number of assets is large. A market with many listed stocks produces a dense dependence matrix that contains thousands of pairwise relationships. Without filtering, it is difficult to identify which relationships are economically meaningful and which stocks occupy central positions in the market structure. Network analysis addresses this issue by representing stocks as nodes and dependence relationships as edges. This allows researchers to examine the topology of market interdependence, including clusters, hubs, bridges, and core-periphery structures.

Financial network analysis is useful because systemic risk depends not only on the strength of individual links, but also on the position of each node within the system. A stock with many strong connections may transmit shocks widely. A stock with high betweenness may act as a bridge between otherwise separate groups. A stock with high eigenvector centrality may be important because it is connected to other important stocks. These centrality concepts help translate dependence relationships into systemic importance. In this sense, network analysis provides an economic interpretation of market structure rather than merely a visual representation of correlations.

Filtered network methods are particularly relevant for financial markets. The Minimum Spanning Tree and Planar Maximally Filtered Graph are common tools for extracting the most informative links from a complete dependence matrix. Tumminello et al. (2005a) introduce the PMFG as a method that retains more information than the Minimum Spanning Tree while preserving a sparse and interpretable network structure. This makes PMFG suitable for analyzing

stock markets because it maintains important clustering patterns without overwhelming the analysis with all possible pairwise links.

Prior studies show that the choice of dependence measure affects the resulting financial network. Correlation networks and tail-dependence networks may reveal different structures because they capture different economic states. Wen, Yang, and W. Zhou (2018) compare Pearson correlation networks with tail-dependence networks of global stock markets using PMFG and find that tail-dependent networks can provide different insights into market influence and risk transmission. This supports the argument that correlation-based networks may not be sufficient for crisis analysis, since extreme downside co-movement can follow a different structure from average co-movement.

The present study builds on this literature by using a network framework to move from pairwise lower-tail dependence to systemic importance. By constructing a PMFG based on lower-tail dependence, the study identifies the strongest downside dependence channels among HOSE-listed stocks. By computing centrality measures and combining them into a systemic importance index, the paper ranks stocks and sectors according to their role in market-wide crash transmission. This approach connects dependence modelling with systemic risk measurement and allows the analysis to answer an economically meaningful question: which parts of the Vietnamese stock market matter most when downside risk becomes systemic?

2.5 Empirical Evidence from Emerging Markets and ASEAN

Emerging markets provide a distinct setting for studying tail dependence and systemic risk. Compared with developed markets, emerging markets often have lower liquidity, higher information asymmetry, greater retail investor participation, and stronger sensitivity to external shocks. These characteristics can make market dependence more unstable and can amplify downside co-movement during crisis periods. In such markets, diversification based on normal-period correlation may be especially unreliable because investors may face abrupt changes in liquidity and risk appetite.

ASEAN markets are particularly relevant for this study because they share several features with Vietnam, including openness to trade, growing financial integration, and vulnerability to global shocks. Duong and Huynh (2020a) examine tail dependence in ASEAN-6 equity markets and find that copula-based methods are useful for capturing dependence structures in the region. Their findings suggest that ASEAN markets exhibit both left- and right-tail dependence, meaning that extreme co-movement is a meaningful feature of regional equity markets. This supports the view that tail dependence should be considered in portfolio allocation and contagion analysis.

Evidence on Vietnam also shows that the market is exposed to external and regional shocks. C. Nguyen, Bhatti, and Henry (2017) document left-tail dependence between the US and Vietnamese stock markets, indicating that Vietnam is vulnerable to joint crash behaviour with the US market during extreme events. T. T. A. Tran (2023) further studies risk spillovers from

the US stock market to Vietnam and shows that external market shocks can affect domestic risk dynamics. These studies imply that Vietnam's stock market cannot be treated as isolated from global financial conditions.

However, much of the existing emerging market and ASEAN literature focuses on cross-country dependence or market-level indices. While this is useful for understanding international contagion, it does not reveal the internal structure of downside dependence within Vietnam. A market-level index can show whether Vietnam is exposed to external shocks, but it cannot identify which domestic stocks or sectors become key transmitters of downside risk. This creates a need for stock-level and sector-level analysis of lower-tail dependence within HOSE.

2.6 Evidence on Systemic Risk, Spillovers, and Connectedness in Vietnam

2.6.1 Systemic Risk in Vietnamese Financial Institutions

Vietnam-specific research has increasingly examined systemic risk and financial connectedness. Vu and D. K. Tran (2019) study systemic risk in the Vietnamese stock market, while P. A. Nguyen, N. P. Nguyen, and H. M. D. Le (2023) develop a Financial Risk Meter approach to assess systemic risk and financial network conditions in Vietnam. These studies show that systemic risk is an important issue in the Vietnamese financial system and that market-based information can be used to monitor periods of heightened vulnerability.

Recent evidence also emphasizes the role of financial networks. A. T. H. Nguyen and T. T. Le (2025) examine financial connectedness among Vietnamese banks, insurance firms, and securities companies using the Diebold and Yilmaz (2012) network framework. They find a consistently high level of interconnectedness among Vietnamese financial institutions, particularly after the COVID-19 pandemic and recent bank runs and market stress. Their abstract reports that banks traditionally serve as primary shock transmitters, while securities companies have become increasingly influential in the post-2020 environment, and insurance firms are relatively isolated net receivers of shocks. The paper also explains that Vietnam is a useful setting because economic growth has outpaced financial development, while the dominance of a few large banks creates concentration risk and potential contagion vulnerabilities.

This evidence is highly relevant for the present study because it confirms that network-based risk transmission matters in Vietnam. However, A. T. H. Nguyen and T. T. Le (2025) focus on financial institutions and volatility connectedness among banks, insurers, and securities firms. The present study differs by examining a broader universe of HOSE-listed stocks across sectors and by focusing on lower-tail dependence rather than general volatility connectedness. Therefore, while the financial connectedness literature provides important motivation, it does not fully answer which stocks and sectors become central to downside crash transmission in the broader equity market.

2.6.2 Sectoral Spillovers in Vietnamese Stock Markets

A second strand of Vietnamese research focuses on sectoral spillovers. Dang, N. T. Nguyen, and Vo (2023) examine dynamic volatility spillovers across Vietnamese sectors and show that intersectoral connectedness varies over time and intensifies during the COVID-19 period. This suggests that Vietnamese sectors are not isolated and that shocks can travel across industries during stress. Similarly, studies of sectoral connectedness show that some industries act as net transmitters of risk while others behave as net receivers, implying that systemic relevance differs across sectors.

Mensi et al. (2024) investigate extreme quantile connectedness between oil and Vietnamese stock sectors. Their findings are important because they show that spillovers can be stronger under extreme market conditions than under normal conditions. This supports the broader argument that average connectedness may not be sufficient for understanding crisis-period transmission. When markets enter bearish or extreme states, the direction and magnitude of spillovers may change substantially.

These sectoral studies are closely related to the present research because they show that risk transmission in Vietnam is heterogeneous across industries. Nevertheless, most sectoral spillover studies use sector indices or aggregate sector-level variables. They therefore cannot identify which individual stocks drive sectoral systemic importance. Moreover, spillover and connectedness frameworks typically focus on volatility or return transmission, rather than lower-tail dependence among stock-level returns. This leaves open the question of whether downside co-movement has a distinct sectoral structure within HOSE.

2.6.3 Copula and Tail Dependence Applications in Vietnam

Copula-based evidence in Vietnam has also developed in recent years. Huynh et al. (2020) examine contagion risks in the Vietnamese banking system using non-parametric and copula approaches. Their study is important because it shows that bank stock returns may exhibit nonlinear dependence and contagion risk that cannot be captured fully by standard correlation measures. C. Nguyen, Bhatti, and Henry (2017) also use copula methods to study tail dependence between Vietnam, China, and the United States, showing that Vietnam is exposed to left-tail dependence with the US market during extreme events.

Other studies apply copulas to portfolio and sectoral questions in Vietnam. For example, Thu Thuy Nguyen, V. C. Nguyen, and Tran Trong Nguyen (2020) examine dependence among oil and gas-related firms, while T. A. Le and Dao (2021) use copula-based approaches for portfolio optimization under downside risk considerations. These studies demonstrate the usefulness of copulas for modelling nonlinear dependence and portfolio risk in Vietnam. However, most existing copula applications focus on either selected sectors, cross-market relationships, or portfolio optimization. They do not construct a stock-level lower-tail dependence network for the whole HOSE market.

Therefore, the Vietnamese literature has made progress in three separate directions: systemic risk measurement, sectoral spillover analysis, and copula-based dependence modelling. The remaining gap is the lack of an integrated framework that combines lower-tail dependence with network-based systemic importance at the stock and sector level. This is the gap addressed by the present study.

2.7 Research Gap and Positioning of the Present Study

The reviewed literature suggests that systemic risk is fundamentally connected to the structure of financial dependence. Prior studies have shown that interconnectedness can support risk sharing in normal periods but can also amplify contagion during stress. The literature also shows that tail dependence is important because crises are characterized by joint extreme losses rather than average co-movement. Copula models provide a suitable framework for capturing nonlinear and asymmetric dependence, while financial networks help translate pairwise dependence into systemic importance.

However, several gaps remain in the context of Vietnam. First, existing studies on Vietnamese financial connectedness mainly focus on financial institutions or sector-level spillovers. They provide important evidence on banks, securities firms, insurance firms, and sectoral indices, but they do not fully identify stock-level channels of downside risk transmission across the broader HOSE market. Second, existing copula-based studies in Vietnam show that tail dependence exists, but they generally do not connect lower-tail dependence to systemic importance rankings of stocks and sectors. Third, correlation-based or volatility-based connectedness measures may not capture the structure of crash co-movement, especially when dependence becomes asymmetric in the lower tail.

The present study addresses these gaps by constructing a lower-tail dependence network among HOSE-listed stocks and comparing it with a conventional Pearson correlation network. The paper contributes to systemic risk measurement by identifying which stocks and sectors become central to downside market transmission. It contributes to dependence modelling by testing whether Clayton lower-tail dependence captures crisis co-movement differently from Pearson correlation. It also contributes to Vietnam-focused financial stability research by examining whether systemic importance is persistent across sectors or changes across crisis regimes.

In this sense, the study is positioned at the intersection of three research streams: tail-dependence modelling, financial network analysis, and systemic risk measurement in emerging markets. Its main contribution is economic rather than purely methodological. It asks whether the Vietnamese stock market contains hidden downside dependence structures that conventional correlation may miss, and whether these structures reveal which sectors are most relevant for crash transmission and macroprudential monitoring.

3 Methodology

3.1 Research Design and Empirical Strategy

This study adopts a market-based network approach to examine lower-tail dependence and sectoral systemic risk among HOSE-listed stocks. The empirical strategy is designed to answer three related questions. First, it measures whether Vietnamese stocks exhibit significant downside co-movement. Second, it identifies which stocks and sectors occupy central positions in the downside dependence network. Third, it evaluates whether lower-tail dependence provides more economically meaningful information than conventional Pearson correlation in explaining crisis-period co-movement.

The methodology uses the Clayton copula to estimate lower-tail dependence among HOSE-listed stocks and to construct a PMFG-based network of downside risk transmission. This framework is appropriate because the research focuses on joint extreme losses, sectoral crash linkages, and systemic importance under market stress, rather than on average co-movement across the full return distribution. Systemic risk becomes economically relevant when losses become synchronized during adverse market conditions, and lower-tail dependence is designed to capture precisely this feature. Pearson correlation is included only as a supporting benchmark to assess whether conventional average linear co-movement captures the same crisis-relevant information as the lower-tail framework. Prior studies show that tail-dependent networks may reveal different structures from correlation networks, particularly during periods of extreme market movements (Wen, Yang, and W. Zhou, 2018; D. Wang and Huang, 2021).

The methodology proceeds in five main stages. First, stock returns are filtered using a GJR-GARCH model to remove stock-specific volatility dynamics. Second, standardized residuals are transformed into pseudo-uniform series and used to estimate Clayton lower-tail dependence. Third, a filtered PMFG network is constructed from the lower-tail dependence matrix. Fourth, network centrality measures are used to build the Systemic Importance Index. Finally, the Pearson correlation benchmark is used in supporting comparisons based on network similarity, ranking comparison, and crisis validation regressions. This design allows the study to assess whether tail-based systemic importance captures downside market vulnerability more effectively than correlation-based importance.

3.2 Data, Sample Construction, and Return Measurement

The study uses daily price data for HOSE-listed stocks over the period 2016–2025. HOSE is selected because it is the main equity trading venue in Vietnam and contains the most liquid and systemically relevant listed firms. Focusing on HOSE also improves data consistency because firms listed on this exchange generally have better trading continuity and disclosure quality than smaller exchanges or less liquid market segments.

The sample is constructed by applying a two-stage filtering procedure to all HOSE-listed

stocks over the January 2016–December 2025 window. First, only stocks with a sufficiently long and continuous trading history are retained, excluding newly listed, thinly traded, or intermittently suspended firms.¹ Second, within each GICS sector, stocks are selected to reflect both economic size and market liquidity, so that the final sample captures firms capable of transmitting shocks in real time rather than merely the largest by capitalisation.² Sector classification is preserved throughout to allow aggregation into sector-level risk measures relevant for macroprudential monitoring. After applying these filters, the final sample consists of 193 stocks spanning 11 GICS sectors, providing a representative and economically meaningful cross-section of the Vietnamese equity market.

Table 3.1: Distribution of Sample Stocks by Sector

Sector	N	%	Tickers
Real Estate	41	21.24	CCL, D2D, DIG, DTA, DXG, EVG, FDC, HAR, HDC, HQC, IJC, ITC, KBC, KDH, KPF, LDG, LGL, LHG, NBB, NHA, NLG, NTC, NTL, NVL, PDR, PTL, QCG, SCR, SGR, SJS, SZL, TCH, TDC, TDH, TIP, TIX, VIC, VPH, VPI, VRC, VRE
Cons. Staples	35	18.13	AAM, ABT, ACL, AFX, AGM, ANV, ASM, BBC, BHN, CLC, CMX, DAT, DBC, FMC, HAG, IDI, KDC, LAF, LIX, LSS, MCH, MSN, NAF, NSC, OGC, PAN, PIT, SAB, SBT, SMB, SSC, TCO, VCF, VHC, VNM
Financials	30	15.54	ACB, AGR, APG, BIC, BID, BMI, BSI, BVH, CTG, CTS, EIB, FTS, HCM, LPB, MBB, MIG, ORS, PGI, SHB, SSI, STB, TVB, TVS, VCB, VCI, VDS, VIB, VIX, VND, VPB
Industrials	26	13.47	BCG, BMP, CII, CTD, CTI, DPG, FCN, GEX, GMD, HAH, HHV, HVN, LCG, PC1, SAM, SCS, SHI, STG, TLG, TMS, TV2, VCG, VGC, VJC, VOS, VSC
Materials	19	9.84	AAA, BFC, CSV, CVT, DCM, DGC, DHC, DPM, DPR, HPG, HSG, HT1, KSB, NHH, NKG, PHR, PTB, SMC, TRC
Utilities	18	9.33	BWE, CHP, CNG, GAS, GEG, HDG, HNA, KHP, KOS, NT2, PGD, PPC, REE, SHP, SJD, TDM, VPD, VSH
Cons. Disc.	12	6.22	CTF, DGW, DRC, GIL, HAX, HHS, MWG, PET, PNJ, TCM, TSC, TTF
Info Tech	5	2.59	CMG, DLG, ELC, FPT, ITD
Energy	4	2.07	PLX, PVD, PVP, PVT
Health Care	2	1.04	DCL, DHG
Media Svc	1	0.52	CTR
Total	193	100.00	

Daily log returns are computed as:

$$r_{i,t} = \ln \left(\frac{P_{i,t}}{P_{i,t-1}} \right) \times 100, \quad (1)$$

¹Specifically, a stock must have been listed no later than 2018 (minimum seven-year listing history) and must have at least 80% non-missing price observations over the full sample window.

²The target number of stocks per sector is set proportionally to each sector's share of total market capitalisation: $n_s = \text{round}(N \times w_s)$, where $w_s = \text{MCAP}_s / \sum_s \text{MCAP}_s$ and N is the total target sample size. Within each sector, stocks are ranked by a composite score $\text{CompositeRank}_i = (\text{Rank}_{\text{MCAP},i} + \text{Rank}_{\text{AvgVolume},i})/2$, and the top n_s stocks with the lowest scores are retained.

where $P_{i,t}$ is the closing price of stock i on day t . Log returns are used because they are time-additive and commonly applied in financial risk modelling. Economically, daily returns reflect how market participants incorporate new information about firm fundamentals, sectoral conditions, liquidity, and macro-financial shocks. Using stock-level returns allows the study to observe how shocks are priced and transmitted across firms within the Vietnamese market.

3.3 Volatility Filtering and Marginal Modelling

Financial return series typically exhibit volatility clustering, fat tails, and asymmetric responses to negative shocks. These features are particularly relevant in emerging markets, where liquidity shocks and investor sentiment can produce abrupt volatility changes. If dependence is estimated directly from raw returns, the resulting dependence measures may be contaminated by stock-specific volatility dynamics. To address this issue, the study first models each stock's marginal return process before estimating cross-stock dependence.

For each stock i , the conditional mean and innovation process are specified as:

$$r_{i,t} = \mu_i + \varepsilon_{i,t}, \quad (2)$$

$$\varepsilon_{i,t} = \sigma_{i,t} z_{i,t}, \quad (3)$$

where μ_i is the conditional mean, $\varepsilon_{i,t}$ is the innovation, $\sigma_{i,t}$ is the conditional volatility, and $z_{i,t}$ is the standardized innovation. The conditional variance is modelled using the GJR-GARCH(1,1) specification proposed by Glosten, Jagannathan, and Runkle (1993a):

$$\sigma_{i,t}^2 = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \gamma_i \varepsilon_{i,t-1}^2 \mathbf{1}_{\{\varepsilon_{i,t-1} < 0\}} + \beta_i \sigma_{i,t-1}^2. \quad (4)$$

In this model, α_i captures the short-run reaction of volatility to past shocks, β_i captures volatility persistence, and γ_i captures the asymmetric or leverage effect. A positive γ_i indicates that negative shocks increase future volatility more strongly than positive shocks of the same magnitude. This is economically meaningful because equity market downturns often generate stronger uncertainty, margin pressure, and liquidity demand than comparable upward movements.

The innovations are assumed to follow a skewed- t distribution. This assumption is appropriate because stock returns often display skewness and excess kurtosis, and skewed- t innovations provide greater flexibility than the normal distribution in capturing fat-tailed return behaviour. GARCH-type filtering before copula estimation is widely used in financial dependence modelling because it separates marginal volatility dynamics from the cross-sectional dependence structure (Patton, 2012; McNeil, Frey, and Embrechts, 2015b).

After estimating the model, standardized residuals are obtained as:

$$\hat{z}_{i,t} = \frac{\hat{\varepsilon}_{i,t}}{\hat{\sigma}_{i,t}}. \quad (5)$$

These standardized residuals represent the return innovations after controlling for time-varying volatility. Economically, this step separates individual volatility risk from systemic dependence risk. This distinction is important because the paper is not primarily interested in whether one stock is volatile on its own, but whether stocks remain connected after their own volatility dynamics have been filtered out.

3.4 Probability Integral Transform and Pseudo-Uniform Series

To estimate copula dependence, the standardized residuals must be transformed into comparable marginal distributions. This study applies a rank-based probability integral transform to obtain pseudo-uniform variables:

$$u_{i,t} = \frac{\text{rank}(\hat{z}_{i,t})}{T + 1}, \quad (6)$$

where T is the number of observations. This transformation maps the standardized residuals into the interval $(0, 1)$. The use of $T + 1$ prevents observations from taking exact boundary values of 0 or 1, which can create numerical problems in copula estimation.

The pseudo-uniform transformation is useful because it allows the dependence structure to be estimated separately from the marginal distributions. This follows the logic of Sklar's theorem, which states that a multivariate joint distribution can be decomposed into its marginal distributions and a copula function (Sklar, 1959; Nelsen, 2006). In practical terms, this means that stocks with different volatility levels, skewness, and return distributions can be compared on the same probabilistic scale. For an emerging market such as Vietnam, where firm-level liquidity and volatility differ substantially across sectors, this step improves the comparability of dependence estimates.

3.5 Dependence Estimation: Correlation versus Lower-Tail Dependence

3.5.1 Pearson Correlation as the Baseline Dependence Measure

The first pipeline estimates Pearson correlation using the standardized residuals $\hat{z}_{i,t}$. For each stock pair (i, j) , the correlation coefficient is computed as:

$$\rho_{ij} = \frac{\sum_t (\hat{z}_{i,t} - \bar{z}_i)(\hat{z}_{j,t} - \bar{z}_j)}{\sqrt{\sum_t (\hat{z}_{i,t} - \bar{z}_i)^2} \sqrt{\sum_t (\hat{z}_{j,t} - \bar{z}_j)^2}}. \quad (7)$$

Pearson correlation is included as a benchmark because it is widely used in portfolio allocation, risk management, and financial network analysis. It captures the average linear co-movement between two assets. From an economic perspective, this corresponds to the

conventional diversification view: assets with lower correlations are expected to provide better risk reduction.

However, Pearson correlation treats upside and downside co-movement symmetrically and summarizes dependence over the full distribution. It may therefore understate crisis-period co-movement if stocks become more strongly connected during extreme downside states. This limitation is central to the present study, because the research question is not simply whether stocks move together on average, but whether they crash together during market stress.

For network construction, the Pearson correlation is converted into a metric distance:

$$d_{ij} = \sqrt{2(1 - \rho_{ij})}. \quad (8)$$

This distance transformation is commonly used in correlation-based financial networks because lower distance corresponds to stronger correlation. The resulting distance matrix forms the basis of the correlation-based PMFG network.

3.5.2 Clayton Copula and Lower-Tail Dependence

The second pipeline estimates lower-tail dependence using the Clayton copula. The bivariate Clayton copula is defined as:

$$C(u_i, u_j; \theta) = \left(u_i^{-\theta} + u_j^{-\theta} - 1 \right)^{-1/\theta}, \quad \theta > 0. \quad (9)$$

The Clayton copula is selected because it captures lower-tail dependence. This property is well suited to the objective of the study, which is to measure joint downside risk rather than general co-movement. In equity markets, lower-tail dependence is economically important because it describes the state in which diversification fails: two stocks are more likely to experience severe losses at the same time. In systemic risk terms, high lower-tail dependence indicates that distress in one stock is more likely to coincide with distress in another.

The Clayton dependence parameter is estimated using the method of moments based on Kendall's tau:

$$\hat{\theta}_{ij} = \frac{2\hat{\tau}_{ij}}{1 - \hat{\tau}_{ij}}, \quad (10)$$

where $\hat{\tau}_{ij}$ is Kendall's tau between $u_{i,t}$ and $u_{j,t}$. The lower-tail dependence coefficient is then computed as:

$$\lambda_{ij}^L = 2^{-1/\hat{\theta}_{ij}}. \quad (11)$$

The coefficient λ_{ij}^L measures the strength of joint downside dependence between stocks i and j . A higher value indicates a greater probability that the two stocks experience extreme negative returns together. If $\hat{\tau}_{ij} \leq 0$, the pair is treated as having no positive lower-tail dependence and λ_{ij}^L is set to zero. This treatment preserves the full pairwise structure while ensuring that only positive downside dependence contributes to the tail network.

This lower-tail dependence pipeline is the main empirical framework of the study. It directly addresses the economic problem that conventional correlation may fail to detect hidden crash linkages. Prior studies show that tail-dependence networks can identify different market structures from correlation networks and may provide more useful information for systemic risk analysis (G.-J. Wang, Xie, and Stanley, 2016; Wen, Yang, and W. Zhou, 2018; D. Wang and Huang, 2021).

3.6 Network Construction Using Planar Maximally Filtered Graphs

After estimating pairwise dependence, the study constructs financial networks using the Planar Maximally Filtered Graph (PMFG). In a complete dependence matrix with N stocks, there are $N(N - 1)/2$ pairwise links. Such a dense structure is difficult to interpret and may obscure the economically meaningful channels of risk transmission. Filtered graph methods reduce this complexity by retaining only the most informative links.

The PMFG is used because it preserves more information than the Minimum Spanning Tree while maintaining a sparse and interpretable structure. For a network with N nodes, the PMFG contains:

$$|E| = 3(N - 2) \quad (12)$$

edges, where E denotes the set of retained links. The planarity condition ensures that the graph can be embedded on a plane without crossing edges. Tumminello et al. (2005b) show that the PMFG is useful for extracting meaningful hierarchical and clustering structures from complex financial systems.

The construction procedure is applied separately to the two dependence matrices. For the Pearson pipeline, stock pairs are sorted by increasing distance d_{ij} , so that the strongest correlations are considered first. For the Clayton pipeline, stock pairs are sorted by decreasing λ_{ij}^L , so that the strongest lower-tail dependence links are considered first. Starting from an empty graph, each candidate edge is added if the resulting graph remains planar. The procedure stops when the graph contains $3(N - 2)$ edges.

The resulting networks are denoted as $PMFG^{corr}$ and $PMFG^{tail}$. The correlation network represents the structure of average co-movement, while the tail network represents the structure of joint downside co-movement. Economically, this distinction allows the study to compare how the Vietnamese market looks under normal dependence versus crash-oriented dependence. If the two networks differ substantially, this would suggest that traditional correlation-based monitoring may miss important channels of downside risk transmission.

3.7 Centrality Measures and Systemic Importance Index

3.7.1 Network Centrality Measures

Once the filtered networks are constructed, the study computes four weighted centrality measures for each stock: weighted degree, betweenness, closeness, and eigenvector centrality. These

measures capture different dimensions of systemic importance.

Weighted degree centrality measures the total strength of a stock's direct connections:

$$C_D^i = \sum_{j \in N_i} w_{ij}, \quad (13)$$

where N_i is the set of neighbouring stocks connected to stock i , and w_{ij} is the edge weight. In the correlation network, w_{ij} is the Pearson correlation. In the tail network, w_{ij} is the lower-tail dependence coefficient. A stock with high weighted degree is directly connected to many strongly related stocks. Economically, this means that its movements are closely tied to a broad set of other stocks.

Betweenness centrality measures the extent to which a stock lies on the shortest paths between other stocks:

$$C_B^i = \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}}, \quad (14)$$

where σ_{st} is the number of shortest paths between stocks s and t , and $\sigma_{st}(i)$ is the number of those paths that pass through stock i . A stock with high betweenness may act as a bridge between different market clusters. In systemic risk terms, such stocks are important because they can connect otherwise separate sectors or groups of firms.

Closeness centrality measures how close a stock is to all other stocks in the network:

$$C_C^i = \frac{N - 1}{\sum_{j \neq i} d(i, j)}, \quad (15)$$

where $d(i, j)$ is the shortest path distance between stocks i and j . In weighted networks, inverse weights are used as distances. A stock with high closeness can reach other stocks through relatively short paths, implying that shocks associated with this stock may diffuse quickly through the network.

Eigenvector centrality measures whether a stock is connected to other important stocks:

$$C_E^i = \frac{1}{\lambda} \sum_{j \in N_i} A_{ij} C_E^j, \quad (16)$$

where A_{ij} is the weighted adjacency matrix and λ is the largest eigenvalue. A stock with high eigenvector centrality may not only have many connections, but also be connected to other systemically relevant stocks. This recursive notion of influence is particularly useful for identifying core stocks in a market network.

Together, these four measures capture complementary aspects of systemic importance: direct connectedness, bridge function, diffusion proximity, and connection to influential nodes. This multi-dimensional view is necessary because systemic relevance cannot be reduced to a single network property. For example, a stock may not have the highest number of direct links but may

still be important because it bridges two major sectors.

3.7.2 Construction of the Systemic Importance Index

To combine the four centrality measures into a single systemic importance measure, the study constructs a Systemic Importance Index (SII) using principal component analysis. Before applying PCA, all centrality measures are normalized to the interval $[0, 1]$. This prevents measures with larger numerical scales from dominating the index.

The SII for stock i is defined as:

$$SII_i = w_D C_D^i + w_B C_B^i + w_C C_C^i + w_E C_E^i, \quad (17)$$

where w_D , w_B , w_C , and w_E are the first principal component loadings associated with weighted degree, betweenness, closeness, and eigenvector centrality, respectively. The first principal component is used because it captures the largest common variation across the centrality measures. This follows the logic of D. Wang and Huang (2021), who combine network centralities through PCA to construct a comprehensive measure of systemic importance in lower-tail dependence networks.

The study constructs two versions of the index:

$$SII_i^{corr} \quad (18)$$

from the Pearson correlation network, and:

$$SII_i^{tail} \quad (19)$$

from the Clayton lower-tail dependence network. The first captures systemic importance under average co-movement, while the second captures systemic importance under downside co-movement. The economic interpretation of the index is therefore different across the two pipelines. SII_i^{corr} identifies stocks that are central in normal linear dependence, while SII_i^{tail} identifies stocks that are central in crash-related dependence.

The tail-based SII is the main measure of systemic importance in this study. A stock with a high SII_i^{tail} is not merely volatile. Rather, it is strongly embedded in the lower-tail dependence network and may therefore contribute to market-wide downside transmission. This interpretation is useful for both investors and regulators. Investors can use it to identify stocks that weaken portfolio diversification during crises, while regulators can use it to monitor stocks or sectors that may amplify market stress.

3.8 Sector-Level Systemic Risk Measurement

Because the study is concerned with sectoral systemic risk, stock-level systemic importance is aggregated to the sector level. For each sector s , the sector-level SII is computed as:

$$SII_s = \frac{1}{N_s} \sum_{i \in s} SII_i, \quad (20)$$

where N_s is the number of stocks in sector s . This aggregation produces a sectoral ranking of systemic importance. The sector-level measure is computed for both the correlation and tail-dependence frameworks, although the main interpretation focuses on SII_s^{tail} .

Sector-level aggregation is important for economic and policy interpretation. In practice, financial instability is often discussed at the sector level, such as banking, real estate, materials, energy, or consumer sectors. A sector with high SII^{tail} may not necessarily have the highest average volatility, but it may be more central to the market's downside dependence structure. This distinction matters because policymakers and market supervisors need to know not only which sectors fall the most, but which sectors are most likely to transmit or amplify losses across the market.

The sector-level analysis also allows the study to evaluate whether systemic importance is persistent or regime-dependent. For example, financial and real estate stocks may be persistently central due to credit and collateral linkages, while materials or energy stocks may become more systemically relevant during trade or commodity-related shocks. This connects the network results to economically meaningful channels of risk transmission.

3.9 Regime-Based Analysis

The analysis is conducted for the full sample and for selected sub-periods representing different market regimes. The purpose of the regime-based analysis is to examine whether lower-tail dependence and systemic importance are stable over time or change with the nature of market stress. This is important because systemic risk is state-dependent. A stock or sector may not be central during normal periods but may become important during a specific shock.

For each regime, the study re-estimates pairwise dependence, reconstructs the Pearson and Clayton PMFG networks, recalculates centrality measures, and recomputes SII^{corr} and SII^{tail} . This produces comparable systemic importance rankings across regimes. The analysis then examines changes in mean lower-tail dependence, sector-level SII rankings, and network topology.

This design allows the study to distinguish between persistent systemic sectors and crisis-specific systemic sectors. Persistent systemic sectors are those that remain central across most regimes, suggesting structural importance in the Vietnamese market. Crisis-specific systemic sectors are those whose importance rises sharply during particular shocks, suggesting exposure to specific economic channels such as trade disruption, pandemic restrictions, liquidity stress,

or tariff-related uncertainty.

3.10 Framework Comparison and Crisis Validation

3.10.1 Network Similarity and Ranking Comparison

To evaluate whether the Pearson and Clayton frameworks identify similar or different market structures, the study computes the Jaccard similarity index between the two PMFG edge sets:

$$J(E^{corr}, E^{tail}) = \frac{|E^{corr} \cap E^{tail}|}{|E^{corr} \cup E^{tail}|}. \quad (21)$$

A high Jaccard value indicates that the two networks retain many common links, while a low value indicates that the correlation and lower-tail frameworks identify different dependence structures. Economically, a low overlap would suggest that the market's average co-movement structure differs from its downside co-movement structure. This would provide evidence that correlation-based risk monitoring may not fully capture crash transmission channels.

The study also compares SII^{corr} and SII^{tail} using Spearman rank correlation. Spearman correlation is appropriate because the main interest is whether the two frameworks rank stocks similarly in terms of systemic importance. Stocks with large rank differences are examined as potential cases of hidden downside risk. These are stocks that may not appear highly systemic under correlation but become important under lower-tail dependence.

3.10.2 Crisis Validation Regression

To assess the economic relevance of the systemic importance measures, the study estimates a cross-sectional crisis validation regression:

$$\bar{R}_i^{crisis} = \alpha + \beta_{tail} SII_i^{tail} + \beta_{corr} SII_i^{corr} + \varepsilon_i, \quad (22)$$

where $\bar{R}_i^{crisis}$ is the average daily return of stock i during a crisis window. Both SII_i^{tail} and SII_i^{corr} are standardized before estimation to make the coefficients comparable. Heteroskedasticity-robust standard errors are used because cross-sectional return dispersion may differ across stocks.

The key coefficient of interest is β_{tail} . If $\beta_{tail} < 0$ and statistically significant, this indicates that stocks with higher tail-based systemic importance experience larger losses during crisis periods. This would support the argument that SII^{tail} has economic relevance beyond network description. In contrast, if β_{corr} is insignificant or weaker, this would suggest that lower-tail dependence provides more useful information about crisis vulnerability than conventional correlation.

This validation step is important because the objective of the study is not only to construct a technically sophisticated network, but to test whether the network captures economically

meaningful downside risk. A systemic importance measure is useful only if it helps explain or anticipate which stocks are most vulnerable when the market enters a stress state.

4 Empirical Results

4.1 Sample Characteristics and Preliminary Diagnostics

Table 4.1 reports the sectoral composition of the final HOSE sample. The final dataset consists of 193 stocks across 11 GICS sectors, after applying the full-window trading-history requirement and the size-liquidity selection rule.³ Real Estate, Consumer Staples and Financials are the three largest sectors in the empirical sample, accounting for 21.24%, 18.13% and 15.54% of selected stocks, respectively. This composition is broadly consistent with the market-capitalization structure of HOSE, where Financials, Real Estate and Consumer Staples account for 37.98%, 26.32% and 11.07% of total sectoral market capitalization, respectively. However, the final stock-count distribution does not fully replicate the original market-capitalization weights, because the empirical design gives priority to data availability, trading continuity and liquidity rather than sector size alone.

Table 4.1: Sample composition by GICS sector

Sector	Number of stocks	Percentage
Real Estate	41	21.24%
Consumer Staples	35	18.13%
Financials	30	15.54%
Industrials	26	13.47%
Materials	19	9.84%
Utilities	18	9.33%
Consumer Discretionary	12	6.22%
Information Technology	5	2.59%
Energy	4	2.07%
Health Care	2	1.04%
Media Services	1	0.52%
Total	193	100.00%

The lower representation of Financials in the final sample is mainly explained by the strict listing-history requirement. Although Financials is the largest HOSE sector by market capitalization (figure 1), several large banks and securities firms are excluded because they were listed or transferred to HOSE after the beginning of the 2016–2025 sample window.⁴ Consequently,

³The initial HOSE universe contains 405 historical symbols, of which 212 are excluded. The main exclusion criteria are insufficient trading history, defined as less than 80% non-missing price observations over the 2016–2025 window, and failure to satisfy the sector-level size-liquidity ranking requirement. This screening rule is necessary because lower-tail dependence estimation requires a sufficiently long and continuous return series across normal and crisis regimes.

⁴In Financials, 15 stocks are excluded, mainly due to insufficient trading history. Notable excluded names include TCB, HDB, TCX, TPB, MSB and OCB. These stocks have substantial market capitalization, with TCB

the Financials sample is concentrated in institutions with longer and more continuous trading records, including the three large state-linked banks VCB, BID and CTG, together with selected private banks, securities companies and insurers. External financing in Vietnam continues to be mediated primarily through the banking sector rather than through long-term capital-market instruments (long-term bond or equity financing). Bank valuations are closely associated with expectations about economic growth, credit expansion, monetary-policy transmission, liquidity conditions and financial-system stability. Therefore, Financials remains one of the dominant sectors in the sample despite the exclusion of several high-market-capitalization banks.

Real Estate also remains strongly represented, with 41 stocks, equivalent to 21.24% of the final sample. This result is consistent with its substantial post-filter market-capitalization share of 26.32% (Figure 1). The large market capitalization of Real Estate in Vietnam reflects the sector's close connection with several structural growth drivers, including urbanization, infrastructure development, industrial-zone expansion and household wealth accumulation. In an emerging market such as Vietnam, real estate assets are not only consumption and investment goods, but also important collateral in the credit system. As a result, listed property developers tend to be valued not only based on current earnings, but also on land banks, project pipelines, expected infrastructure spillovers and the perceived long-term appreciation of urban and industrial land. The sectoral weight is also influenced by the presence of large listed groups. In particular, VIC accounts for 20.74% and VRE for 1.23% of market capitalization in the updated sectoral structure. The role of VIC is especially relevant at the end of 2025, when VIC, VHM, VPL and VRE were reported to contribute more than 19 points to the VN-Index.⁵⁶ Therefore, the strong representation of Real Estate in the final sample reflects both its market size and its structural relevance to downside-risk transmission, particularly through the credit-collateral linkage between property developers, banks and investor sentiment.

Consumer Staples ranks second in the final sample by stock count, with 35 stocks, equivalent to 18.13%, despite accounting for a lower market-capitalization share of 11.07% in the sectoral structure. This difference indicates that Consumer Staples contains a relatively broad set of firms that satisfy the history and liquidity requirements, compared with sectors where market capitalization is more concentrated in a smaller number of large stocks. The sector includes large and liquid consumer, food, beverage and agricultural firms such as VNM, MSN, SAB, MCH,

around VND 236 trillion, HDB around VND 132 trillion, TCX around VND 115 trillion, TPB around VND 45 trillion, MSB around VND 40 trillion and OCB around VND 30 trillion in the updated result summary (by 2025). Their exclusion is therefore methodological rather than economic: the filter preserves full-cycle comparability for tail-risk estimation.

⁵During the same period, VIC closed at around VND 169,600 per share and increased by approximately seven times relative to the beginning of the year. At the end of 2025, VIC's strong performance was linked to improved investor sentiment after Vingroup simultaneously commenced several large-scale projects. This development likely strengthened expectations about future project execution, infrastructure expansion and ecosystem-based cash-flow generation, even before these prospects were fully reflected in short-term earnings.

⁶In Real Estate, 14 stocks are excluded, also mainly because of the history filter. Notable excluded names include VHM, BCM, KHG, CRE and DXS. The exclusion of VHM is particularly important because it is a large-cap real estate stock, but it does not satisfy the full 2016–2025 trading-history requirement used in the study.

KDC, VHC, PAN and SBT, together with smaller firms that maintain sufficient trading continuity over the sample period. Its relatively high representation is therefore not solely determined by aggregate market capitalization, but also by the availability of continuous stock-level data across the full 2016–2025 window. Consumer Staples is relevant for an emerging-market downside-risk analysis because the sector combines defensive consumption characteristics with exposure to commodity prices, export demand, agricultural supply chains and domestic purchasing power. Its inclusion therefore broadens the empirical network beyond the Financials–Real Estate core and improves the sectoral coverage of lower-tail dependence on HOSE.

The relevance of Financials and Real Estates is further supported by the credit-collateral linkage between real estate and the banking system. Real estate assets are widely used as collateral, accounting for around 70% of total collateral during 2024 and 2025 ⁷, while property developers rely on bank loans, corporate bond issuance, and secondary-market liquidity as important financing channels. This relationship creates a transmission mechanism through which changes in interest rates, credit conditions, or collateral values may affect both sectors simultaneously. In the equity market, investors often view Financials and Real Estate as indicators of domestic credit and asset-price conditions. This interpretation is consistent with recent empirical evidence on systemic risk in Vietnam. A. T. H. Nguyen and T. T. Le (2025) show that interconnectedness among Vietnamese financial institutions increases during periods of market stress and bank runs, with banks acting as primary shock transmitters and securities firms functioning as amplifiers of systemic risk. In addition, Loan, Hai, and Nga (2024) highlight the role of the real estate sector in transmitting stock crash effects across the Vietnamese market, while Dang, N. T. Nguyen, and Vo (2023) identify Real Estate as the sector absorbing the largest magnitude of volatility spillovers during turbulent periods. Accordingly, the high representation of Financials and Real Estate in the dataset is not only a feature of sample composition, but also supports the study’s objective of capturing the network structure of systemic downside-risk transmission and contagion dynamics in Vietnam.

Table 4.2 reports the descriptive statistics of daily logarithmic returns across GICS sectors. Average daily returns are positive across all sectors, ranging from 0.0515% in Energy to 0.1151% in Media Services. However, the interpretation of sectoral mean returns should be cautious for sectors with relatively small sample sizes, such as Media Services and Health Care. Among the larger sectors, Financials records the highest average daily return at 0.0835%, followed by Materials at 0.0780% and Real Estate at 0.0720%. Since the present study focuses on downside-risk transmission rather than average return performance, the distributional characteristics of returns provide more relevant evidence for the subsequent tail-dependence analysis.

The volatility pattern reveals clear cross-sectoral differences. Real Estate records the highest standard deviation, at 2.7390%, followed by Consumer Discretionary at 2.5909% and Consumer Staples at 2.5359%. Financials and Industrials also display relatively high volatility, with

⁷In 2025, BIDV 70%, CTG 71% and VCB 70%

standard deviations of 2.4257% and 2.4363%, respectively. By contrast, Utilities has the lowest standard deviation, at 1.8912%, followed by Health Care at 2.1224%. This pattern suggests that the sectors with the largest market representation are not homogeneous in terms of risk structure. Real Estate combines a high sectoral market-capitalization share with significant return dispersion, acting as a highly vulnerable sector that receives and heavily transmits extreme market shocks (Dang, N. T. Nguyen, and Vo, 2023; Loan, Hai, and Nga, 2024). Concurrently, while Financials may exhibit lower unconditional volatility than Real Estate, the sector remains the primary catalyst for systemic risk (A. T. H. Nguyen and T. T. Le, 2025). This profound systemic relevance stems directly from its central node in credit intermediation and liquidity transmission across the economy (A. T. H. Nguyen and T. T. Le, 2025). Crucially, this dynamic is amplified by a strong structural linkage with the property market, wherein banks heavily exposed to real-estate developers via corporate bond holdings or direct lending are highly susceptible to joint tail crashes during credit downturns.

Within Financials, the sectoral average masks a clear contrast between securities firms and banks. The most volatile financial stocks in the sample are securities companies, including APG, VIX, ORS, AGR, VDS, CTS, FTS, BSI, VCI and HCM, with daily standard deviations ranging from approximately 2.65% to 3.01%. This reflects the business model of securities firms, whose earnings are more directly exposed to market turnover, proprietary trading performance and margin lending conditions. In contrast, large banks display lower return volatility, with VCB, ACB and EIB recording standard deviations of 1.61%, 1.75% and 1.81%, respectively. This distinction is important for the interpretation of systemic risk. Banks provide the core credit-intermediation function, while securities firms may amplify market stress through their direct exposure to trading liquidity and margin-financing cycles.

Consumer Staples also requires a sector-specific interpretation. Although this sector is often considered defensive in developed markets, the Vietnamese Consumer Staples group contains a broad set of agricultural, aquaculture, food-processing and trading firms. The most volatile stocks in this sector include PIT, TCO, DAT, CMX, LAF, OGC, SSC, ANV, AGM and HAG, with daily standard deviations ranging from approximately 2.89% to 3.37%. These firms are more exposed to commodity-price movements, export demand, weather-related production risk and firm-specific liquidity conditions. At the same time, the sector also includes large and more stable consumer firms such as VNM, SAB and MSN, with daily standard deviations of 1.48%, 1.77% and 2.14%, respectively. Therefore, the relatively high representation of Consumer Staples in the sample does not imply a uniform defensive profile. Rather, it reflects an internally heterogeneous sector that combines stable large-cap consumer stocks with more cyclical agricultural and export-oriented firms.

Real Estate records the highest sector-level standard deviation, which is consistent with its sensitivity to credit conditions, collateral values, infrastructure expectations and investor risk appetite. The highest-volatility real estate stocks include DTA, QCG, PTL, LDG, VRC,

NHA, HAR, EVG, FDC and TIX, with daily standard deviations ranging from approximately 3.05% to 3.56%. These names are mostly mid-cap or smaller property-related firms, whose prices tend to be more sensitive to liquidity conditions, project-specific news and speculative trading. By contrast, larger real estate stocks such as VRE, KDH, VIC and NVL record lower standard deviations of 2.14%, 2.16%, 2.21% and 2.45%, respectively. This intra-sectoral contrast indicates that Real Estate's high volatility is not driven solely by large-cap firms, but also by smaller and more liquidity-sensitive developers. This feature is relevant for the later network analysis, because distress in volatile property stocks may transmit through investor sentiment, collateral values and margin-financing channels.

The higher-order moments further indicate that sectoral returns deviate from the Gaussian assumption. Skewness values are relatively moderate across most sectors, ranging from -0.0207 in Energy to 0.4934 in Health Care. Most sectors display positive skewness, suggesting that the return distributions are slightly right-skewed over the sample period. More importantly, all sectors exhibit positive excess kurtosis, indicating fat-tailed return distributions. Utilities records the highest excess kurtosis, at 3.5345, followed by Health Care at 2.7212, Consumer Staples at 1.7224, Energy at 1.6778 and Industrials at 1.6544. Real Estate, despite having the highest volatility, records a lower excess kurtosis of 1.1334. This suggests that Real Estate risk is reflected more through consistently higher return dispersion than through the highest tail thickness among sectors.

The Jarque–Bera statistics are large across all sectors, leading to the rejection of the null hypothesis of normality. This result indicates that non-normal return behavior is a broad feature of HOSE sectoral returns rather than a sector-specific phenomenon. The coexistence of positive excess kurtosis and significant non-normality is consistent with empirical evidence from emerging equity markets, where return distributions often display fat tails, asymmetric adjustment and stronger dependence during periods of market stress (Duong and Huynh, 2020b; V. C. Nguyen and Thu Thuy Nguyen, 2022; Z. Wang et al., 2022).

These descriptive results provide direct motivation for the empirical strategy of the study. First, the presence of fat-tailed return distributions supports the use of GJR-GARCH filtering to account for time-varying volatility and asymmetric return dynamics at the individual-stock level. Second, the rejection of normality suggests that conventional linear correlation may be insufficient for capturing downside co-movement. Since systemic risk is primarily associated with joint adverse movements rather than average return co-movement, the Clayton copula framework is appropriate for estimating lower-tail dependence after transforming standardized residuals into pseudo-uniform variables through the Probability Integral Transform. Therefore, the descriptive statistics provide preliminary evidence in favor of a tail-sensitive approach to systemic downside-risk transmission on HOSE.

To account for these distributional properties, each stock return series is filtered using a

Table 4.2: Descriptive statistics of daily log returns by GICS sector

Sector	Obs.	Mean	Std. dev.	Min	Max	Skewness	Exc. kurtosis	JB stat.
Consumer Discretionary	29633	0.0650	2.5909	-7.2464	7.3171	0.0897	1.3739	2368.8
Consumer Staples	86200	0.0541	2.5359	-7.4594	7.5000	0.1087	1.7224	10822.7
Energy	9230	0.0515	2.3395	-7.3113	7.4387	-0.0207	1.6778	1081.3
Financials	72445	0.0835	2.4257	-7.5000	7.4951	0.0706	1.6337	8114.9
Health Care	4998	0.0797	2.1224	-7.0000	7.0064	0.4934	2.7212	1740.4
Industrials	63287	0.0681	2.4363	-7.4775	7.4903	0.1185	1.6544	7363.4
Information Technology	12488	0.0730	2.4829	-7.1111	7.1429	0.1726	1.4777	1196.5
Materials	46942	0.0780	2.3760	-7.4933	7.4074	0.0574	1.6103	5095.6
Media Services	2010	0.1151	2.3866	-7.3718	7.3926	0.0885	1.4693	181.8
Real Estate	100145	0.0720	2.7390	-7.4349	7.4919	0.0940	1.1334	5506.7
Utilities	42996	0.0630	1.8912	-7.4496	7.4951	0.1611	3.5345	22560.1

univariate GJR-GARCH(1,1) model with skewed- t innovations before the dependence structure is estimated. This procedure serves two purposes. First, it models stock-specific conditional volatility dynamics, including volatility clustering and heavy-tailed innovations. Second, it produces standardized innovations that can be used to estimate lower-tail connectedness after controlling for firm-level volatility persistence (Li and Rose, 2009b). The GJR-GARCH specification is appropriate because it allows conditional volatility to respond asymmetrically to positive and negative shocks, consistent with the leverage effect documented by Glosten, Jagannathan, and Runkle (1993b). From a systemic-risk perspective, filtering returns in this way helps reduce the possibility that dependence estimates are driven by unmodelled heteroskedasticity rather than genuine downside co-movement among stocks.

Table 4.3: GJR-GARCH persistence and asymmetric volatility response by GICS sector

Sector	Median α	Median γ	Median β	Persistence	$\alpha + \gamma$	$(\alpha + \gamma)/\alpha$
Consumer Discretionary	0.1355	0.0306	0.8141	0.9649	0.1661	1.26
Consumer Staples	0.2328	0.0685	0.7237	0.9908	0.3014	1.45
Energy	0.1015	0.0879	0.8426	0.9880	0.1894	1.52
Financials	0.1368	0.0768	0.8139	0.9891	0.2136	1.44
Health Care	0.2702	0.0574	0.7011	1.0000	0.3276	1.25
Industrials	0.1358	0.0536	0.8245	0.9871	0.1894	1.33
Information Technology	0.2166	0.0334	0.7577	0.9910	0.2500	1.22
Materials	0.1087	0.0813	0.8457	0.9950	0.1900	1.79
Media Services	0.1224	0.2545	0.7503	1.0000	0.3769	3.08
Real Estate	0.1577	0.0487	0.8119	0.9940	0.2064	1.25
Utilities	0.2237	0.0478	0.7576	1.0052	0.2715	1.15

Table 4.3 reports the median GJR-GARCH parameters by sector, together with the negative-to-positive volatility response ratio, measured as $(\alpha + \gamma)/\alpha$. The persistence estimates are high across all sectors, ranging from 0.9649 in Consumer Discretionary to 1.0052 in Utilities. Most sectors record persistence close to unity, including Materials at 0.9950, Real Estate at 0.9940, Information Technology at 0.9910, Consumer Staples at 0.9908 and Financials at 0.9891. This indicates that volatility shocks in HOSE-listed stocks tend to decay slowly, which is consistent with the presence of volatility clustering in emerging equity markets. The high

persistence estimates support the use of a conditional volatility filter before estimating cross-stock dependence, since raw returns may otherwise contain sector-specific heteroskedasticity that can contaminate the dependence measures.

The asymmetric volatility response is also evident across sectors. Media Services records the highest asymmetry ratio, at 3.08, but this result should be interpreted cautiously because the sector contains only one stock in the final sample. Among the broader sectors, Materials has the highest asymmetry ratio at 1.79, followed by Energy at 1.52, Consumer Staples at 1.45 and Financials at 1.44. These values indicate that negative shocks are associated with larger conditional-variance responses than positive shocks of comparable size. For Materials, the relatively high asymmetry ratio is consistent with the sector's exposure to commodity-price declines, construction demand, operating leverage and inventory-cycle risk. Firms in steel, chemicals, cement, rubber and related input industries may experience faster revisions in expected revenues and margins when output prices or downstream demand weaken, while fixed production costs and working-capital commitments adjust more slowly.

For Energy, the asymmetry ratio of 1.52 reflects the sector's exposure to downside risks related to oil and gas price movements, policy changes, geopolitical conditions and capital-intensive project risk. Since energy firms often require large fixed investment and may be sensitive to balance-sheet conditions, adverse price or regulatory shocks can generate stronger volatility responses than favorable news of similar magnitude. Consumer Staples also records a relatively high asymmetry ratio of 1.45. This suggests that the sector should not be interpreted as uniformly defensive in the Vietnamese context. The sample includes not only large consumer firms, but also agricultural, aquaculture, food-processing and trading firms that are exposed to commodity prices, export demand, input-cost volatility and weather-related production risks. These characteristics can make negative shocks more consequential for conditional volatility.

Financials records an asymmetry ratio of 1.44, indicating that adverse shocks generate a stronger volatility response than positive shocks. This is consistent with the sector's role in credit intermediation, margin financing, market liquidity and investor sentiment. Within Financials, securities firms are particularly sensitive to market turnover, proprietary trading conditions and margin-lending cycles, while banks are more directly linked to credit growth, asset quality and provisioning risk. Therefore, the sector's asymmetric volatility response reflects both market-liquidity channels and credit-risk channels. This result is relevant for the subsequent network analysis because financial institutions can function as transmitters or amplifiers of downside shocks, especially when adverse market movements coincide with tightening liquidity conditions.

Real Estate displays an asymmetry ratio of 1.25. Although this is lower than the ratios observed for Materials, Energy, Consumer Staples and Financials, it should not be interpreted as evidence of limited systemic relevance. The sector records high volatility in the descriptive statistics and remains structurally connected to the financial system through bank credit, refi-

nancing conditions, collateral values and policy-sensitive project approval. Negative shocks, such as tighter credit conditions, bond-market stress or delays in project implementation, may weaken expected cash flows and reduce collateral capacity. These mechanisms can transmit stress to Financials through the credit-collateral channel examined in the subsequent lower-tail dependence analysis.

By contrast, Utilities records the lowest asymmetry ratio, at 1.15, followed by Information Technology at 1.22. The lower ratio in Utilities suggests that negative and positive shocks produce more comparable conditional-variance responses, despite the sector's high volatility persistence. This may reflect the more stable demand profile of electricity, water and infrastructure-related services. For Information Technology, the relatively low asymmetry ratio is consistent with the sector's comparatively asset-light characteristics and lower direct exposure to commodity-price declines or collateral-based financing. However, both sectors still display persistent volatility dynamics, indicating that they should not be excluded from the dependence analysis.

This pattern is consistent with the broader literature on systemic risk and tail dependence. Recent tail-risk network studies show that systemic fragility may be underestimated when dependence is measured only through average co-movement, as extreme downside events can generate nonlinear and asymmetric linkages across assets (D. Wang and Huang, 2021; Z. Wang et al., 2022). For Vietnam, prior studies also document that market connectedness and downside-risk transmission increase during periods of stress, including the COVID-19 episode and financial-sector turbulence (V. C. Nguyen and Thu Thuy Nguyen, 2022; A. T. H. Nguyen and T. T. Le, 2025). The diagnostic results in this paper therefore provide an empirical basis for the subsequent copula and network analysis. The return data exhibit fat tails, persistent volatility and asymmetric shock responses, which support the use of lower-tail dependence as a relevant measure of systemic downside risk beyond average linear correlation.

Overall, the preliminary evidence supports the empirical strategy of the study. The descriptive statistics reveal non-normal return behavior, while the GJR-GARCH estimates indicate strong volatility persistence and sectoral heterogeneity in asymmetric shock responses. Methodologically, these stylized facts support the extraction of standardized innovations and the application of probability integral transforms for copula calibration. More importantly, the evidence suggests that systemic risk on HOSE should be assessed through extreme downside connectedness rather than relying solely on unconditional volatility or standard linear correlation measures.

4.2 Lower-Tail Dependence and Sectoral Downside Co-movement

Table 4.4 reports the estimated Clayton lower-tail dependence coefficients for the full sample and across market regimes. In the full sample, the mean and median lower-tail dependence coefficients are 0.1273 and 0.0994, respectively, while the 90th percentile reaches 0.2991. These

Table 4.4: Summary of Clayton lower-tail dependence coefficients

Regime	Mean λ_L	Median λ_L	90th pct.	Std. dev.
Full Sample	0.1273	0.0994	0.2991	0.1230
Normal	0.1418	0.1170	0.3264	0.1320
Trade War	0.1058	0.0341	0.3238	0.1435
COVID-19	0.2103	0.1611	0.5185	0.2048
Bond Shock	0.3815	0.4156	0.6992	0.2533
Trump Tariff	0.6106	0.6943	0.8874	0.2776

values indicate that downside connectedness exists among HOSE-listed stocks, but that the typical pairwise lower-tail dependence remains moderate when the entire 2016–2025 period is pooled together. This is consistent with the interpretation that the full-sample estimate represents an average dependence structure across heterogeneous market states, including normal periods, gradual corrections, sharp crisis episodes and recovery phases. The full-sample estimates should therefore be interpreted as a baseline measure of latent downside connectedness rather than a complete characterization of systemic crash risk. Since systemic risk is inherently state-dependent, the regime-specific estimates provide more direct evidence on how dependence changes when market stress becomes concentrated. This distinction is important because a market-wide drawdown does not necessarily imply uniformly high lower-tail dependence. Rather, λ_L captures the extent to which stocks move jointly in the lower tail after stock-specific conditional volatility has been filtered.

Figure 2 complements Table 4.4 by comparing the evolution of lower-tail dependence with market breadth conditions during crisis regimes.

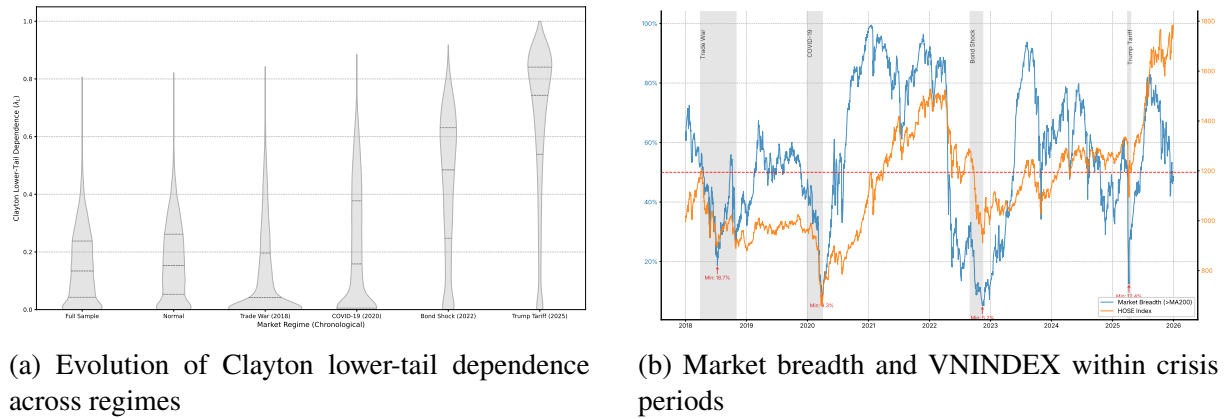


Figure 2: Lower-tail dependence and market breadth across crisis regimes

The regime-level results show substantial variation in lower-tail dependence across shock periods. During normal periods, the mean and median coefficients are 0.1418 and 0.1170, respectively. During the Trade War period, both measures decline to 0.1058 and 0.0341. This suggests that the 2018 shock generated weaker joint lower-tail dependence than the normal benchmark, despite the VN-Index falling from 1196.61 to 914.76, equivalent to a decline of 23.55%. The market-breadth evidence supports this interpretation. During the Trade War

window, the share of stocks trading above their 200-day moving average ranged from 18.65% to 56.48%, with an average of 38.51%. This indicates that downside pressure was present but relatively uneven across stocks, consistent with a more gradual and heterogeneous external shock.

By contrast, COVID-19 is associated with a higher degree of lower-tail dependence. The mean λ_L increases to 0.2103, while the median rises to 0.1611 and the 90th percentile reaches 0.5185. The VN-Index declined from 966.67 to 662.53 during this period, corresponding to a drawdown of 31.46%. Market breadth also deteriorated substantially, with the share of stocks above their 200-day moving average falling to a minimum of 9.33% and averaging 30.54%. Compared with the Trade War episode, the COVID-19 period therefore reflects a broader and more synchronized deterioration in market conditions. This is consistent with the nature of the pandemic shock, which affected domestic mobility, consumption, production activity, investor sentiment and liquidity conditions at the same time.

The Bond Shock regime exhibits a much stronger increase in lower-tail dependence. The mean coefficient reaches 0.3815, the median rises to 0.4156 and the 90th percentile reaches 0.6992. Although the VN-Index drawdown, from 1277.35 to 942.90, or 26.18%, is smaller than the COVID-19 decline, the dependence structure is considerably more concentrated in the lower tail. This difference is also reflected in market breadth. During the Bond Shock window, the share of stocks above their 200-day moving average fell to a minimum of 5.18% and averaged only 14.81%, the weakest breadth condition among the four shock periods. This suggests that the Bond Shock was not only a market correction, but also a broad-based deterioration in participation and trend strength. Economically, this is consistent with a domestic funding shock in which stress in the corporate bond market, refinancing constraints and weakening confidence in leveraged real-estate firms transmitted pressure to financial institutions and liquidity-sensitive stocks.

The Trump Tariff episode records the highest lower-tail dependence among all regimes. The mean λ_L reaches 0.6106, the median increases to 0.6943 and the 90th percentile rises to 0.8874. This is notable because the VN-Index decline during the event window is smaller than in the other crisis periods, falling from 1317.33 to 1219.12, or 7.46%. The sharp increase in lower-tail dependence therefore cannot be explained by the size of the index drawdown alone. Instead, it indicates a rapid synchronization of downside movements across stocks. The market-breadth statistics support this interpretation: the share of stocks above their 200-day moving average dropped to a minimum of 12.44%, with an average of 29.29%, within a short event window. This pattern suggests that the episode operated as a sudden external repricing shock, where uncertainty over trade policy triggered a cross-sector risk-off response and a rapid deterioration in market participation.

Taken together, these results indicate that lower-tail dependence is regime-dependent and cannot be inferred solely from the magnitude of market decline. The Trade War period involved

a sizeable VN-Index correction, but lower-tail dependence remained comparatively weak, suggesting that the shock was absorbed unevenly across firms. COVID-19 generated a stronger increase in joint downside dependence, consistent with a broad macroeconomic disruption. The Bond Shock produced a sharper deterioration in market breadth and a much higher median λ_L , reflecting the domestic credit and refinancing channel. The Trump Tariff episode generated the highest tail dependence despite the smallest index drawdown, indicating that sudden repricing shocks can produce highly synchronized downside movements even over short windows.

These findings support the use of a lower-tail dependence framework for systemic-risk analysis on HOSE. Standard market indicators, such as index returns, capture the aggregate price decline, while market breadth captures the extent of participation deterioration. Clayton lower-tail dependence adds a further dimension by identifying whether losses become jointly concentrated across stocks. The evidence therefore suggests that systemic downside risk is not determined only by how far the market falls, but also by how broadly and how synchronously downside movements spread through the equity network. This finding is consistent with recent tail-risk and systemic-risk studies, which emphasize that extreme dependence is regime-dependent and tends to intensify during periods of financial stress (D. Wang and Huang, 2021; Z. Wang et al., 2022; A. T. H. Nguyen and T. T. Le, 2025).

Table 4.5 reports the strongest intra-sector lower-tail dependent pairs across the full sample and crisis regimes. The results show that internal downside co-movement is concentrated primarily in Financials and Materials, with one notable Consumer Staples pair. In the full sample, the leading pairs are HCM–SSI, HSG–NKG, SSI–VND, ASM–IDI and DCM–DPM, all with lower-tail dependence coefficients above 0.70. Financials account for six of the top ten intra-sector pairs, indicating that downside co-movement within this sector is not confined to a single subsector, but extends across both securities firms and banks.

Table 4.5: Top intra-sector lower-tail dependent pairs

Pair	Sector	Full Sample	Normal	Trade War	COVID-19	Bond Shock	Trump Tariff
HCM–SSI	Financials	0.782	0.805	0.770	0.829	0.804	0.914
HSG–NKG	Materials	0.757	0.796	0.565	0.657	0.899	0.927
SSI–VND	Financials	0.752	0.765	0.781	0.687	0.865	0.961
ASM–IDI	Consumer Staples	0.739	0.730	0.700	0.813	0.835	0.914
DCM–DPM	Materials	0.716	0.774	0.494	0.645	0.899	0.873
ACB–MBB	Financials	0.708	0.703	0.819	0.840	0.736	0.927
HCM–VCI	Financials	0.699	0.738	0.577	0.642	0.860	0.873
HCM–VND	Financials	0.697	0.719	0.704	0.670	0.747	0.857
CTG–MBB	Financials	0.696	0.693	0.831	0.719	0.841	0.873
SSI–VCI	Financials	0.692	0.733	0.612	0.497	0.845	0.805

The concentration of Financials in the intra-sector ranking is consistent with the structure of Vietnam’s financial market. Securities-firm pairs such as HCM–SSI, SSI–VND, HCM–VCI, HCM–VND and SSI–VCI form a persistent internal transmission cluster within Financials. This pattern is consistent with the business model of Vietnamese securities companies, whose main revenue sources are highly sensitive to market conditions. Brokerage income depends on market

turnover, margin-lending income depends on investor leverage demand and collateral conditions, proprietary trading is exposed to mark-to-market portfolio performance, and investment-banking revenue varies with the IPO, issuance and M&A cycle. Although the relative importance of these activities differs across firms, the underlying exposure is similar. SSI and HCM are closely linked to brokerage, margin lending and proprietary trading; VND is sensitive to the retail trading and margin cycle; while VCI has substantial exposure to proprietary investment and more cyclical investment-banking activity. During market downturns, lower trading value, declining margin demand, forced deleveraging, mark-to-market losses and weaker capital-raising activity can occur simultaneously. These common revenue sensitivities explain why securities firms form a strong lower-tail dependence cluster within the financial sector. Banking pairs such as ACB–MBB and CTG–MBB also display strong lower-tail dependence. This pattern is noteworthy because these banks differ in their credit profiles. ACB is more closely associated with retail and SME lending, CTG has historically maintained a stronger position in large corporate and state-linked lending, while MBB has distinct exposure to selected corporate and project-related segments. Hence, their lower-tail dependence is unlikely to arise solely from identical loan-portfolio structures. A more plausible interpretation is that these pairs reflect system-wide banking stress, including funding-cost pressure, interbank-market liquidity conditions, deposit competition, credit-growth constraints and broad investor confidence toward the banking sector. This channel is examined further in the regime-level analysis, where domestic credit stress may generate joint downside movements across banks even when their credit structures differ.

Materials constitute the second major source of intra-sector downside co-movement. The steel pair HSG–NKG and the fertilizer pair DCM–DPM represent two distinct commodity-linked channels. HSG and NKG share exposure to hot-rolled coil input costs, construction demand, real-estate activity, export orders and trade-policy risks. When demand weakens or selling prices decline faster than input costs, margin pressure can occur across the steel segment, leading to joint downside movements. DCM and DPM are both closely linked to the fertilizer cycle, with earnings affected by urea prices, gas input costs, regional supply-demand conditions and agricultural demand. These common industry fundamentals help explain why Materials pairs remain strongly connected in the lower tail, especially during stress regimes.

The Consumer Staples pair ASM–IDI also appears among the strongest intra-sector pairs. This result is economically intuitive because IDI is closely connected to the Sao Mai Group ecosystem and is commonly described as a core member in ASM-related aquaculture, seafood-processing and export activities. Hence, the lower-tail dependence between ASM and IDI likely reflects both shared exposure to pangasius-sector fundamentals and group-level business linkages. Shocks to export demand, fish prices, input costs, working-capital conditions or investor sentiment toward the group may therefore affect both stocks simultaneously. Although Consumer Staples is often treated as a defensive sector, this pair illustrates that some Vietnamese consumer-staples stocks are exposed to export demand, commodity prices, working-capital conditions and supply-chain disruptions. The sector therefore contains both stable large-cap

consumer names and more cyclical agriculture- or export-oriented firms.

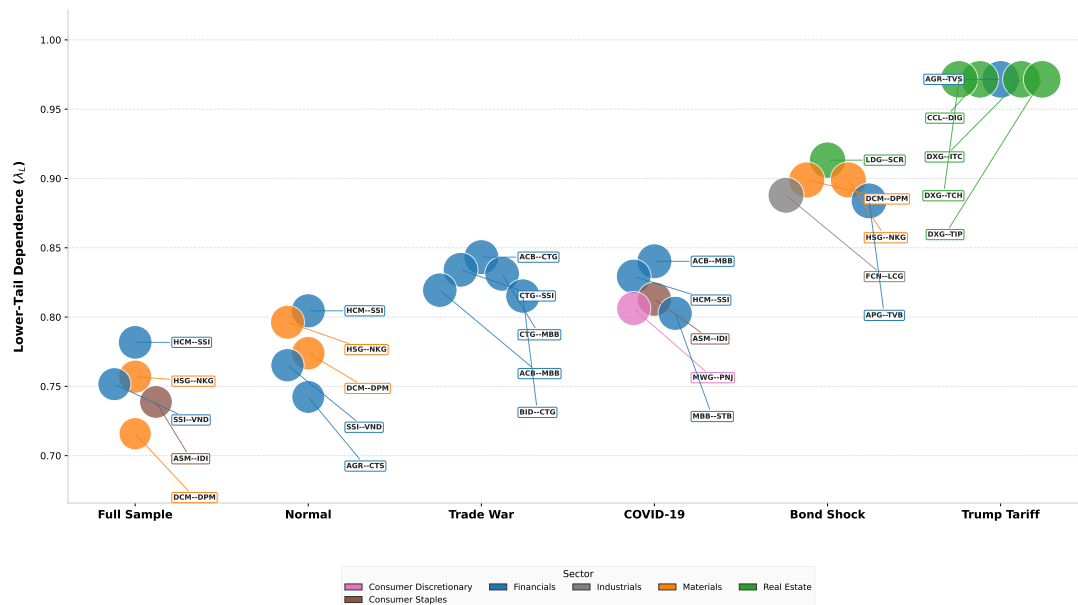


Figure 3: Strongest intra-sector lower-tail dependent pairs

Figure 3 further illustrates the regime-dependent nature of intra-sector lower-tail dependence. During the Trade War period, the dominant pairs are concentrated mainly in Financials, particularly among banks and financial institutions. Since the different credit profiles of ACB, CTG and MBB have already been noted, their joint appearance in this regime is better interpreted as evidence of system-wide market congestion risk rather than identical balance-sheet exposure. In this context, financial stocks may have been repriced together as investors reassessed domestic liquidity, interbank-market conditions, funding-cost pressure and confidence toward the banking system. COVID-19 displays a more diversified structure, combining Financials with consumer-facing firms such as MWG–PNJ and ASM–IDI. This is consistent with the nature of the pandemic shock, which affected both liquidity conditions and operating activity. The MWG–PNJ link reflects a common retail-disruption channel, as both firms were exposed to temporary store closures and the shift toward online sales, although MWG had partial support from essential-goods retail while PNJ was more exposed to non-essential jewelry demand.

The Bond Shock and Trump Tariff episodes show two different forms of crisis-specific concentration. During the Bond Shock, the strongest intra-sector links span Real Estate, Materials, Industrials and Financials, suggesting that domestic funding stress affected several credit-sensitive and liquidity-sensitive sectors simultaneously. Real Estate pairs such as LDG–SCR reflect the sensitivity of leveraged property stocks to refinancing pressure, confidence deterioration and margin-related liquidity withdrawal. The presence of DCM–DPM is also economically consistent, as both firms are major gas-based urea producers and are jointly exposed to urea prices, gas input costs and agricultural demand. In risk-off periods, these firms may be traded as a common commodity-cycle pair. By contrast, the Trump Tariff episode is dominated by Real Estate pairs such as CCL–DIG, DXG–ITC, DXG–TCH and DXG–TIP. This suggests that the

external tariff shock triggered a rapid repricing of high-beta property stocks, especially those with relatively strong downside co-movement during market downtrends.

Overall, the updated intra-sector evidence shows that lower-tail dependence is not evenly distributed across the market. Financials form the most persistent internal transmission cluster, especially through securities-firm pairs and selected banking pairs. Materials display strong commodity-cycle dependence, particularly through steel and fertilizer pairs. Consumer Staples also contains important tail-dependent pairs when firms share exposure to agriculture, aquaculture, export demand and working-capital risk. These findings support the argument that systemic downside risk on HOSE is shaped by both financial linkages and sector-specific economic fundamentals, rather than by aggregate market declines alone.

Table 4.6 reports the strongest cross-sector lower-tail dependent pairs. The results suggest that downside co-movement across sectoral boundaries is concentrated around several economically meaningful transmission channels. The Utilities–Energy channel, represented by GAS–PVD and GAS–PLX, reflects common exposure to energy-sector sentiment, oil and gas price expectations, inflation risk and broad risk-off selling in energy-related stocks. GAS is more closely linked to gas infrastructure and midstream activities, PVD is exposed to drilling and upstream service demand, while PLX reflects petroleum distribution and fuel-market conditions. Therefore, these links are better interpreted as energy-theme repricing channels rather than direct earnings linkages. In addition, pairs such as KSB–LCG indicate a Materials–Industrials linkage associated with construction materials, infrastructure activity and contractor-related demand, suggesting that lower-tail dependence can also emerge between real-economy sectors sharing exposure to investment cycles.

The Real Estate–Financials channel is especially important, as shown by DXG–SSI, DXG–VND and DXG–HCM. This channel connects property developers with securities firms, combining credit-collateral risk and market-liquidity risk. DXG is sensitive to housing-market liquidity, legal progress, refinancing conditions and investor sentiment toward real estate assets, while SSI, VND and HCM are exposed to trading value, margin lending, proprietary investment and capital-market sentiment. When real estate stress increases, declining collateral values, margin calls, forced selling and weaker market turnover can affect both property and securities stocks simultaneously. A related Materials–Financials channel also appears through HPG–SSI and HCM–HPG. HPG is linked to steel prices, construction demand, infrastructure expectations and the property cycle, while securities firms are sensitive to market liquidity and margin activity. This indicates that real-economy cyclicity and financial-market leverage can reinforce each other during downside episodes.

Overall, lower-tail dependence in the Vietnamese equity market is moderate in the pooled sample, but becomes more pronounced across specific stress regimes and sectoral clusters. The pooled estimates reveal a baseline layer of latent downside connectedness, while the regime-specific results show stronger dependence during funding, liquidity and rapid repricing shocks.

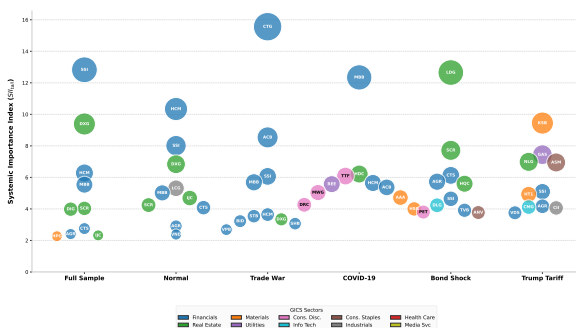
Table 4.6: Top cross-sector lower-tail dependent pairs

Pair	Sector 1	Sector 2	Full Sample	Normal	Trade War	COVID-19	Bond Shock	Trump Tariff
HHS–TCH	Consumer Discretionary	Real Estate	0.606	0.682	0.142	0.501	0.890	0.857
DLG–HQC	Information Technology	Real Estate	0.576	0.624	0.318	0.642	0.918	0.927
GAS–PVD	Utilities	Energy	0.575	0.572	0.640	0.707	0.375	0.950
HPG–SSI	Materials	Financials	0.546	0.556	0.649	0.641	0.595	0.927
GAS–PLX	Utilities	Energy	0.546	0.523	0.685	0.536	0.494	0.914
DXG–SSI	Real Estate	Financials	0.539	0.496	0.693	0.668	0.770	0.972
KSB–LCG	Materials	Industrials	0.527	0.586	0.013	0.517	0.819	0.901
DXG–VND	Real Estate	Financials	0.524	0.505	0.601	0.506	0.783	0.927
DXG–HCM	Real Estate	Financials	0.523	0.501	0.614	0.764	0.756	0.927
HCM–HPG	Financials	Materials	0.522	0.527	0.574	0.561	0.576	0.927

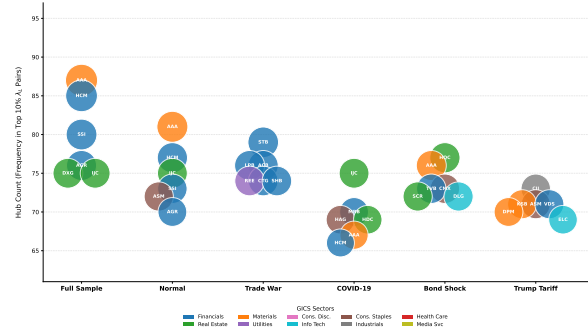
At the sectoral level, the structure of dependence is shaped by both internal sectoral fundamentals and cross-sector financial linkages. Financials and Real Estate form a credit-collateral and liquidity-transmission channel; Materials, Energy and Utilities reflect commodity, infrastructure and industrial-cycle exposure; and selected high-beta pairs across sectors capture common sensitivity to investor sentiment and liquidity withdrawal. These findings indicate that systemic downside risk on HOSE is a regime-dependent and sectorally structured phenomenon, rather than a constant level of average market co-movement.

4.3 Systemic Importance and Regime-Dependent Crash Transmission

This subsection examines whether systemic importance in the lower-tail network is stable over time or varies across market regimes. Two complementary indicators are used. The first is the Clayton-based systemic importance index, SI_{tail} , which measures each stock's structural centrality within the lower-tail PMFG. The second is hub frequency, defined as the number of times a stock appears in the top 10% of pairwise lower-tail dependence links. While SI_{tail} captures a stock's position in the crash-transmission network, hub frequency captures its repeated participation in the strongest co-crash relationships. These measures should not be interpreted simply as rankings of firm size, firm quality or standalone risk. Instead, they indicate how the market historically priced certain stocks as focal points of downside-risk transmission under different regimes.



(a) Top SI_{tail} stocks across market regimes



(b) Systemic hub tickers by frequency in top 10% lower-tail dependence pairs

Figure 4: Systemic importance and hub frequency across market regimes

Figure 4a shows that, in the full sample, systemic importance is concentrated around Finan-

cials and Real Estate. The leading nodes include securities firms, banks and real-estate stocks, particularly SSI, DXG, HCM, MBB and SCR. This pattern indicates that the Financials–Real Estate nexus is not only a crisis-period phenomenon, but also a persistent feature of the lower-tail network. The economic interpretation is that the market has historically treated these stocks as connectors between credit conditions, collateral values, market liquidity and investor leverage. This is particularly important because Financials and Real Estate can transmit stress beyond their own sectors through both balance-sheet linkages and investor-behaviour channels. During periods of stress, investors often shift from return optimization to risk preservation. Financial stocks are then interpreted as observable indicators of system-wide liquidity and confidence, while real-estate stocks are treated as signals of collateral values, refinancing capacity and asset-price stability. As a result, declines in banks, securities firms or property developers may lead investors to reassess risk across a wider set of sectors, even when the initial shock is sector-specific. Securities firms are central because their earnings and balance-sheet risks are closely linked to trading turnover, margin lending, proprietary trading and forced-selling pressure. Banks reinforce the credit and liquidity channel because stress in the banking sector can be interpreted as a signal of tighter funding conditions, weaker credit growth or deteriorating asset quality. Real-estate firms are central because their valuations depend on refinancing conditions, bond-market confidence, project liquidity and collateral values. When real-estate sentiment deteriorates, investors may extend the concern from property developers to banks, materials, construction, consumer demand and market intermediaries through a narrative-contagion mechanism. Therefore, the Financials–Real Estate nexus becomes a powerful source of downside-risk transmission because it combines credit intermediation, collateral valuation, market liquidity, leverage constraints and investor de-risking behaviour.

After establishing this full-sample structure, the regime-specific rankings indicate that systemic importance rotates when the source of stress changes. During COVID-19, systemic importance broadens beyond the Financials–Real Estate core. The appearance of MBB and HCM reflects the continued importance of credit conditions and securities-market liquidity, while HDC, REE and MWG point to real-estate activity, infrastructure-linked demand, utilities and consumer-facing business disruptions. This broader composition is consistent with COVID-19 as an economy-wide shock. The pandemic affected credit demand, asset-quality expectations, household consumption, store operations, supply chains and market liquidity simultaneously. Therefore, the lower-tail network became more cross-sectoral because market expectations linked several parts of the real economy and financial system to the same adverse shock.

The Bond Shock provides clearer evidence of a domestic credit-collateral transmission regime. Systemic importance becomes concentrated around leveraged real-estate and securities-related stocks, with LDG, SCR, HQC and selected securities firms playing prominent roles. This pattern is consistent with an episode driven by weaker corporate bond-market confidence, refinancing pressure and concerns over real-estate balance sheets. In this regime, market

expectations appear to have connected property developers, collateral values, funding access and liquidity-sensitive financial intermediaries into one transmission channel. As refinancing risk increased and collateral values weakened, stress could be transmitted from property developers to securities firms and then to the broader market through margin pressure, forced selling and declining market turnover. The Bond Shock therefore reflects a setting in which credit stress and asset-price declines became mutually reinforcing.

The Trump Tariff regime displays a different form of rotation. Systemic importance shifts toward materials, energy, real estate, consumer staples and financial intermediaries. KSB and HT1 represent construction-material and infrastructure-related cyclicalities, GAS captures energy and commodity sentiment, while NLG and CII reflect the sensitivity of real-estate and infrastructure-linked stocks to external repricing and risk-off behaviour. ASM also indicates that agriculture- or export-linked consumer-staples firms can become systemically relevant when trade-policy uncertainty affects market expectations. Financial intermediaries remain present because external repricing shocks can reduce market turnover, increase margin pressure and trigger liquidity-sensitive selling. This regime therefore suggests that sudden external shocks may rotate systemic importance away from the domestic credit-collateral core and toward stocks exposed to trade uncertainty, commodity sentiment and high-beta market de-risking.

Taken together, the SII_{tail} rankings show that systemic importance is not fixed across regimes. The full sample is anchored by the Financials–Real Estate nexus, reflecting the persistent role of credit, collateral, securities-market liquidity and investor leverage in Vietnam. COVID-19 broadens systemic importance across financial, real-estate, consumer, utility and materials-related stocks because the shock affected both financial conditions and real economic activity. The Bond Shock concentrates systemic importance in property and securities-related stocks, consistent with a domestic funding and refinancing shock. The Trump Tariff episode rotates systemic importance toward externally sensitive and high-beta sectors, including materials, energy, real estate, consumer staples and financial intermediaries. These patterns suggest that the market’s perception of systemic stocks changes with the origin of stress, even though Financials and Real Estate remain recurrent components of the lower-tail network.

Figure 4b provides additional evidence on the persistence and rotation of systemic hubs across regimes. In the full sample and normal periods, recurrent hubs such as AAA, HCM, SSI and IJC suggest that latent downside connectivity was anchored around materials-cycle exposure, securities-market liquidity and real-estate sensitivity. This complements the SII_{tail} results because hub frequency captures repeated pairwise co-crash participation, while SII_{tail} captures structural centrality in the filtered network. During the Bond Shock, hub leadership shifts toward stocks associated with real-estate stress, commodity-linked activity and liquidity-sensitive financial transmission. During the Trump Tariff episode, hubs such as CII, ASM and KSB indicate that sudden external repricing shifted repeated co-crash linkages toward infrastructure-linked, agriculture- or consumer-staples-related and construction-material stocks. Thus, hub

frequency reinforces the interpretation that systemic relevance reflects market expectations about transmission channels rather than firm size alone.

Overall, the joint evidence from SHI_{tail} and hub frequency indicates that the stocks and sectors contributing most to market-wide downside risk on HOSE are both persistent and state-contingent. Financials and Real Estate form the stable domestic transmission core because they connect credit provision, collateral values, margin lending, market liquidity and investor leverage. However, crash transmission does not remain confined to this core. COVID-19 expands systemic importance across sectors because the shock affected consumption, credit, property activity, utilities, production and liquidity simultaneously. The Bond Shock concentrates systemic importance in the Real Estate–Financials nexus through refinancing pressure, collateral-value deterioration and forced-selling risk. The Trump Tariff episode rotates systemic importance toward Materials, Energy, Consumer Staples, infrastructure-linked firms and liquidity-sensitive Financials through external trade repricing, commodity sentiment and market-wide de-risking. Therefore, systemic importance on HOSE is best understood as a regime-dependent network property: Financials and Real Estate are the persistent core, while Materials, Energy, Consumer Staples and infrastructure-related stocks become systemically important when the dominant shock changes the market’s expectations about future cash flows, liquidity and risk-bearing capacity.

4.4 Correlation versus Lower-Tail Dependence: Network Divergence and Crisis Validation

This subsection evaluates whether lower-tail dependence provides additional information on downside co-movement relative to conventional Pearson correlation. Conceptually, the two measures capture different aspects of dependence. Pearson correlation summarizes average linear co-movement over the full return distribution, whereas copula-based lower-tail dependence focuses on the probability that two assets experience extreme negative outcomes together. This distinction is relevant because financial dependence can be asymmetric and state-dependent, particularly during market downturns (Patton, 2012; McNeil, Frey, and Embrechts, 2015b). Network studies also suggest that tail-dependence networks may reveal systemic-risk structures that are not fully captured by correlation-based networks (Wen, Yang, and W. Zhou, 2018; D. Wang and Huang, 2021).

Table 4.7 shows a clear divergence between the Pearson-based and Clayton-based networks. The Jaccard similarity remains low across all crisis regimes, ranging from 0.125 during COVID-19 to 0.248 during the Bond Shock. This indicates that the two frameworks recover substantially different network structures during market stress. In COVID-19 and the Trump Tariff episode, the overlap is particularly low, at 0.125 and 0.130, respectively, implying that most lower-tail connections are not captured by the Pearson-based PMFG. Even during the Bond Shock, where the Jaccard value is relatively higher, more than three quarters of the lower-tail network

Table 4.7: Network Divergence and Crisis Validation

Regime	Jaccard	β_{tail}	p_{tail}	β_{corr}	p_{corr}	R^2
Trade War	0.196	-0.048	0.24317	0.000	0.98792	0.082
COVID-19	0.125	-0.125	0.00484	0.011	0.81306	0.080
Bond Shock	0.248	-0.421	0.01330	0.066	0.51531	0.342
Trump Tariff	0.130	-0.167	0.00005	-0.096	0.01591	0.084

Notes: Jaccard measures the overlap between Pearson and Clayton PMFG edges. Crisis validation regressions relate each stock's average daily log return during the corresponding crisis window to SII_{tail} and SII_{corr} . Both regressors are cross-sectionally standardized. Lower Jaccard values indicate greater divergence between correlation and lower-tail dependence networks. Negative and significant β_{tail} indicates that stocks with higher lower-tail systemic importance experience larger crash losses.

edges remain outside the correlation-based network. These results suggest that average linear co-movement and downside co-crash dependence represent different dimensions of market connectedness.

This divergence is consistent with the conceptual difference between the two measures. Pearson correlation captures co-movement across the full return distribution and may therefore be influenced by normal-period observations. By contrast, the Clayton lower-tail network focuses on joint downside realizations, which are more closely related to liquidity pressure, margin deleveraging, forced selling, funding stress and collateral deterioration. The low network overlap during COVID-19 and the Trump Tariff episode suggests that crisis conditions can reorganize market dependence away from ordinary correlation patterns and toward state-contingent lower-tail linkages.

The crisis validation regressions provide further support for the lower-tail framework. During the Bond Shock, SII_{tail} is negative and statistically significant, with $\beta_{tail} = -0.421$, while SII_{corr} is statistically insignificant. This regime also records the highest explanatory power, with $R^2 = 0.342$, indicating that lower-tail centrality is particularly informative in a domestic funding and collateral-stress episode. The result suggests that stocks located closer to the lower-tail transmission core experienced larger realized losses, whereas correlation-based systemic importance did not explain the cross-sectional crash pattern with comparable strength.

A similar but weaker pattern appears during COVID-19, where SII_{tail} remains negative and statistically significant, while SII_{corr} is insignificant. This indicates that lower-tail systemic importance captured vulnerabilities that became relevant when the pandemic shock affected both financial conditions and real economic activity. During the Trump Tariff episode, both measures are significant, but the lower-tail coefficient is larger in absolute value, with $\beta_{tail} = -0.167$ compared with $\beta_{corr} = -0.096$. This suggests that the tariff shock involved both broad market co-movement and lower-tail crash dependence, with the Clayton-based measure providing the stronger signal of downside vulnerability. In contrast, neither measure is significant during the Trade War, consistent with the interpretation that this episode was more gradual and heterogeneous across firms.

Overall, the evidence indicates that Pearson correlation and Clayton lower-tail dependence capture different aspects of systemic connectedness. Pearson correlation measures average co-movement, whereas lower-tail dependence identifies stocks that become jointly vulnerable in extreme downside states. The low Jaccard similarity confirms that the two network structures diverge during crisis regimes. More importantly, the crisis validation results show that SII_{tail} has stronger and more consistent explanatory power for realized crash losses than SII_{corr} . This supports the interpretation that lower-tail network centrality is more informative for systemic crash-risk analysis, particularly when market stress is driven by funding pressure, collateral deterioration, margin deleveraging, liquidity withdrawal or rapid external repricing.

4.5 Summary of Findings

This study provides three main findings. First, lower-tail dependence is a meaningful feature of the Vietnamese equity market. The results show that HOSE-listed stocks are not only connected through average return co-movement, but also through joint downside risk. This dependence is strongly regime-dependent, becoming more pronounced during acute stress episodes such as the Bond Shock and the U.S. reciprocal tariff shock. This suggests that downside co-movement intensifies when the market faces liquidity deterioration, funding pressure, or sudden external repricing.

Second, the structure of lower-tail dependence is economically organized rather than random. The strongest dependence patterns are concentrated in sectors such as Financials, Real Estate, Energy, and Materials. These results point to two major transmission channels. The first is the Financials–Real Estate credit-collateral channel, where banks, securities firms, and real-estate stocks are jointly exposed to liquidity conditions, collateral values, margin lending, and refinancing pressure. The second is the Materials–Energy industrial channel, where firms share exposure to commodity prices, construction demand, input costs, and investment-cycle expectations. Therefore, lower-tail dependence reflects identifiable economic linkages behind crash transmission in Vietnam’s stock market.

Third, systemic importance is both persistent and regime-dependent. Financials and Real Estate form the stable core of the lower-tail network, confirming their central role in transmitting downside risk across HOSE. However, the identity of the most systemically important stocks changes across crisis regimes. Domestic funding stress tends to strengthen the Real Estate–Financials nexus, while external shocks shift systemic importance toward Materials, Energy, and liquidity-sensitive financial stocks. This indicates that systemic importance is not a fixed firm characteristic, but depends on the nature of the shock and the transmission channel activated during each period.

Finally, the comparison between Pearson correlation and Clayton lower-tail dependence shows that conventional correlation does not fully capture crash-transmission structures. The two networks display limited overlap, indicating that average co-movement and downside co-

movement identify different forms of market connectedness. More importantly, the crisis validation results show that SII_{tail} explains realized crisis-period losses more consistently than SII_{corr} . This supports the view that lower-tail dependence provides a more informative measure of systemic downside risk than conventional correlation.

Taken together, the findings answer the three research questions of this study. HOSE-listed stocks exhibit meaningful lower-tail dependence, this dependence is concentrated around economically important sectoral channels, and lower-tail copula networks provide a clearer picture of crash co-movement than Pearson correlation. Overall, the evidence suggests that systemic risk in Vietnam's stock market is better understood as a regime-dependent lower-tail network, rather than as a structure based solely on average linear co-movement.

5 Discussion and Conclusion

5.1 Discussion of Key Findings

The findings of this study suggest that systemic risk in the Vietnamese equity market is more effectively captured through lower-tail dependence than through average return co-movement. While Pearson correlation measures the general tendency of stock returns to move together, it does not directly capture whether stocks experience joint extreme losses. This distinction is important because systemic risk becomes most relevant during adverse market conditions, when diversification benefits weaken and losses become more concentrated across assets. The evidence of widespread positive lower-tail dependence among HOSE-listed stocks indicates that downside risk is not purely firm-specific. Instead, it is transmitted through a broader market structure that becomes more visible during stress periods. This finding is consistent with the view that financial dependence is often asymmetric and stronger in adverse states, especially in emerging markets where liquidity conditions are thinner and investor sentiment can adjust rapidly (Duong and Huynh, 2020b; Z. Wang et al., 2022; McNeil, Frey, and Embrechts, 2015a).

The results further show that lower-tail dependence is not randomly distributed across the market. Rather, it is concentrated around economically meaningful sectoral linkages, particularly Financials, Real Estate, Energy, and Materials. Among these sectors, Financials and Real Estate form the most persistent core of downside-risk transmission. This pattern is economically relevant in the Vietnamese context because banks, securities firms, and real-estate companies are closely connected through credit conditions, collateral values, refinancing pressure, margin lending, and investor risk appetite. When stress emerges in the real estate sector, concerns over collateral quality, corporate bond repayment capacity, and financial-sector exposure may increase simultaneously. Conversely, stress in the financial sector can reduce liquidity provision and amplify selling pressure in asset-heavy sectors. This finding is broadly consistent with prior studies showing that systemic risk and financial connectedness in Vietnam are strongly influenced by interactions among banks, securities firms, and other financial institutions (Vu

and D. K. Tran, 2019; P. A. Nguyen, N. P. Nguyen, and H. M. D. Le, 2023; A. T. H. Nguyen and T. T. Le, 2025).

In addition to the Financials–Real Estate nexus, the results identify an important Materials–Energy channel. Stocks in these sectors are connected through shared exposure to commodity prices, construction demand, input costs, and investment-cycle expectations. During periods of external repricing or macroeconomic uncertainty, these common exposures may cause firms in Materials and Energy to experience stronger joint downside movements. This finding suggests that lower-tail dependence reflects not only financial linkages, but also real-sector transmission channels. Therefore, crash transmission in Vietnam’s stock market should be understood as a combination of credit-collateral mechanisms, liquidity conditions, and industrial linkages across cyclically sensitive sectors.

Another important finding is that systemic importance is both persistent and regime-dependent. Financials and Real Estate remain structurally important across several regimes, confirming their central role in the lower-tail network. However, the identity of the most systemically important stocks changes depending on the nature of each shock. Domestic funding stress tends to strengthen the Real Estate–Financials nexus, while external repricing shocks increase the importance of Materials, Energy, and liquidity-sensitive financial stocks. This indicates that systemic importance should not be interpreted as a fixed characteristic of individual firms. Rather, a stock becomes systemically important when its sectoral exposure corresponds to the dominant transmission channel of a specific crisis regime. This result supports the broader systemic-risk literature, which emphasizes that financial fragility is shaped not only by firm size, but also by network position, interconnectedness, and exposure to common shocks (Adrian and Brunnermeier, 2016; Acharya et al., 2017b; D. Wang and Huang, 2021).

The comparison between Pearson correlation and Clayton lower-tail dependence further confirms that average co-movement and crash co-movement are distinct concepts. The limited overlap between the two PMFG networks indicates that correlation-based networks do not fully recover many of the links identified by the lower-tail network. More importantly, the crisis validation results show that SII_{tail} explains realized crisis-period losses more consistently than SII_{corr} . This finding supports the methodological choice of using a lower-tail copula approach, since copulas allow the dependence structure to be modeled separately from marginal return distributions and are particularly suitable when financial returns exhibit non-normality, asymmetry, and tail dependence (Nelsen, 2006; Joe, 2014). Overall, the evidence suggests that lower-tail networks provide a more informative representation of crash-transmission risk in Vietnam’s stock market than conventional correlation networks.

5.2 Implications

The findings have important implications for market supervisors, policymakers, and investors. For regulators and market supervisors, the results suggest that systemic-risk monitoring should

not rely only on volatility, liquidity, or correlation-based indicators. Although these measures remain useful, they may understate market fragility when downside dependence is concentrated in specific sectors or when co-crash relationships intensify during stress periods. A lower-tail network indicator such as SHI_{tail} can complement existing monitoring tools by identifying stocks and sectors that are repeatedly involved in joint downside movements. This is particularly relevant for Vietnam, where financial-sector connectedness, real-estate exposure, and market liquidity conditions can jointly affect overall market stability (Huong and Hoai, 2021; A. T. H. Nguyen and T. T. Le, 2025).

The evidence also suggests that regulators should pay close attention to the Financials–Real Estate nexus. Since this nexus acts as a structural core of crash transmission, stress in these sectors may have implications beyond the directly affected firms. During periods of credit tightening, corporate bond-market stress, declining collateral values, or margin pressure, regulators may benefit from monitoring not only individual firm-level risk indicators, but also the position of firms within the lower-tail dependence network. Such monitoring can help identify potential transmission channels before losses become more widespread across the market.

For investors and portfolio managers, the results indicate that diversification strategies based solely on Pearson correlation may provide an incomplete assessment of portfolio risk. Stocks that appear only moderately connected under normal market conditions may still experience strong joint downside movements during crises. Therefore, portfolio construction should consider not only average return co-movement, but also lower-tail dependence among assets. This is especially important for portfolios with large exposure to Financials, Real Estate, Materials, or Energy, as these sectors are shown to play important roles in downside-risk transmission. From a risk-management perspective, incorporating lower-tail dependence can help investors evaluate potential losses under stress conditions and reduce the risk of overestimating diversification benefits (Markowitz, 1952; Low, 2015; T. A. Le and Dao, 2021).

The findings also have implications for the interpretation of systemic importance. Since systemic importance changes across regimes, investors and supervisors should avoid relying only on static measures such as firm size, market capitalization, or normal-period correlation. A stock may become more systemically relevant when its sectoral exposure is aligned with the dominant shock affecting the market. For example, real-estate and financial stocks may become more central during domestic liquidity or funding stress, while materials and energy stocks may become more important during external repricing or commodity-related shocks. This regime-dependent perspective provides a more flexible framework for understanding how systemic risk evolves over time.

5.3 Limitations and Future Research

This study has several limitations. First, the sample focuses on HOSE-listed stocks, which represent a major part of Vietnam’s equity market but do not fully capture the entire listed market.

Future research could extend the analysis to HNX and UPCoM to provide a broader picture of systemic risk across Vietnam's stock exchanges. Such an extension would be useful because market liquidity, firm characteristics, and investor composition may differ across exchanges.

Second, this study applies pairwise Clayton copulas to capture lower-tail dependence. This approach is suitable for identifying asymmetric downside co-movement between pairs of stocks. However, it does not fully capture higher-dimensional dependence among multiple stocks. Future research could apply vine copulas or other multivariate dependence models to examine more complex transmission structures. These methods may provide additional insight into whether systemic risk is transmitted through bilateral links, sectoral clusters, or broader multi-asset dependence structures.

Third, the analysis is conducted at the regime level. This design allows the study to compare lower-tail dependence across normal and crisis periods, but it may not fully capture the continuous evolution of systemic risk. Future studies could apply time-varying copulas or dynamic network models to trace how lower-tail dependence changes before, during, and after stress episodes. This would be particularly useful for identifying early-warning signals of market fragility and for examining whether systemic-risk indicators rise before realized market losses occur.

Fourth, although this study interprets sectoral transmission channels using economic reasoning, it does not directly incorporate firm-level balance-sheet information, ownership networks, corporate bond exposure, or margin-lending data. Future research could combine market-based lower-tail networks with firm-level financial data to better explain why certain stocks become systemically important during specific stress regimes. For example, information on leverage, debt maturity, collateral exposure, and institutional ownership could help clarify the mechanisms behind the Financials–Real Estate credit-collateral channel.

These limitations do not invalidate the findings, but they define the scope within which the results should be interpreted. The study provides evidence that lower-tail dependence is useful for understanding crash transmission in Vietnam's stock market. Future research can extend this framework by using broader samples, richer firm-level data, and more dynamic modeling approaches.

5.4 Conclusion

This study examines systemic risk and lower-tail dependence in Vietnam's stock market using a copula-based network approach. The results show that HOSE-listed stocks are connected not only through average return co-movement, but also through joint downside risk. Lower-tail dependence becomes stronger during stress periods and is concentrated around economically meaningful sectoral channels, especially the Financials–Real Estate nexus and the Materials–Energy channel. These findings indicate that crash transmission in Vietnam's stock market is shaped by both financial linkages and real-sector exposures.

The study also shows that systemic importance is regime-dependent. Financials and Real Estate form a persistent core of the lower-tail network, but the most systemically important stocks vary across crisis regimes. Domestic funding stress tends to strengthen the Real Estate–Financials channel, while external shocks shift systemic importance toward Materials, Energy, and liquidity-sensitive financial stocks. This suggests that systemic importance should be understood as a dynamic outcome of network position, sectoral exposure, and crisis-specific transmission channels, rather than as a fixed firm characteristic.

Finally, the comparison between Pearson correlation and Clayton lower-tail dependence demonstrates that conventional correlation does not fully capture crash-transmission structures. The lower-tail network identifies different links and provides stronger explanatory power for realized crisis-period losses. Overall, the findings support the central argument of this study: systemic risk in Vietnam’s stock market is not adequately captured by average linear co-movement, but is better understood as a regime-dependent lower-tail network shaped by sectoral linkages, liquidity conditions, and crisis-specific transmission channels.

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